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The aspartate-to-homoserine pathway recurs across maize grain amino acids, soil nitrogen, and yield selection

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Abstract

Grain nitrogen in maize depends on nitrogen uptake, assimilation, and allocation, but it is unclear whether the same metabolic functions underlie natural variation in grain amino acid composition, genetic variation associated with soil nitrogen, and response to selection. We tested this in three populations, each with a distinct nitrogen-related phenotype. In the Goodman–Buckler Association Panel, grain free amino acids were dominated by asparagine and proline and were highly heritable, and genome-wide association mapping of 71 traits implicated genes for amino acid metabolism and transport. The loci shared across the most traits lay near a peptidase A1 gene and endoplasmic reticulum membrane protein genes. In 3,107 georeferenced landraces from across the Americas, association mapping of soil nitrogen at collection sites identified variants near an aspartate kinase isozyme and an omega-amidase that passed tests for confounding with geography and climate. Indian Chief and Jarvis responded to 14 cycles of reciprocal recurrent selection for yield with different changes in nitrogen allocation. The two populations diverged genetically, and Indian Chief accumulated more free amino acids, mostly proline, while protein-bound amino acids were unchanged. Their candidate regions included protein-turnover genes in Indian Chief and amino acid importers in Jarvis, consistent with these population-specific phenotypes. Each of the four analyses highlighted a different enzyme of the aspartate-to-homoserine pathway, the only pathway that remained significant after correction across all pathways. Together, these results point to aspartate-family metabolism as a shared source of candidate genes for how nitrogen is partitioned among grain amino acid pools in maize.

Keywords maize, grain nitrogen, free amino acids, aspartate-family amino acids, genome-wide association, reciprocal recurrent selection

Introduction

Nitrogen (N) is essential for maize growth and kernel development, but its use in agriculture is a major sustainability challenge. Crop production relies heavily on N fertilizer, yet much of the applied N is not recovered by the crop and is lost through leaching, runoff, or gaseous emissions (Ojeda-Rivera et al., 2025). Reducing these losses requires understanding how plants acquire, assimilate, transport, and remobilize N, and how they allocate it to the developing grain (Masclaux-Daubresse et al., 2010a; Havé et al., 2017). In maize kernels, N is held in free amino acids (FAAs) and protein-bound amino acids (PBAAs), so grain amino acid composition is a direct biochemical readout of how N is allocated to the grain.

Plant N metabolism centers on a few key amino acids. Ammonium is assimilated via the GS/GOGAT cycle, producing glutamine and regenerating glutamate from 2-oxoglutarate, so Glu/Gln metabolism links N assimilation with carbon metabolism (Masclaux-Daubresse et al., 2010a; Hodges, 2002). Glutamine and asparagine are the main N-rich transport and storage forms, while glutamate and aspartate donate amino groups to carbon skeletons (Bernard and Habash, 2009; Masclaux-Daubresse et al., 2010a). Aspartate is particularly important because it feeds asparagine and the aspartate-family amino acids (methionine, threonine, lysine, isoleucine) (Viola, 2001; Wang et al., 2018; Han et al., 2021). During grain filling, protein degradation in senescing leaves releases amino acids that supply grain N (Havé et al., 2017; Liu et al., 2008). Proline adds a stress-related component, with genotype-dependent accumulation and export from maize leaves under water stress (Carceller et al., 1999). Thus, shifts in grain FAA composition may reflect redistribution among multiple N pools, not just changes in total N.

Most grain N is stored in storage proteins, so protein-bound amino acid composition is largely determined by the proteins produced. The free amino acid pool is small but reflects amino acids delivered to

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and metabolized in the developing kernel, and may therefore better indicate N supply. How much this pool varies heritably among lines, and which genes control that variation, is less well known than for protein-bound amino acids (Shrestha et al., 2022).

Maize germplasm enables grain N metabolism to be studied in three ways. First, standing variation among inbred lines. The Goodman–Buckler panel captures much of the allelic diversity in public breeding and has been used to map grain composition traits (Flint-Garcia et al., 2005; Shrestha et al., 2022). Second, adaptation to soil. The landraces evolved under local conditions across the Americas, so alleles affecting N acquisition or use may correlate with soil N at collection sites. Georeferenced landraces enable environmental GWAS (Romero Navarro et al., 2017), but soil N covaries with geography and climate and must be controlled. Third, selection for yield. Fourteen cycles of reciprocal recurrent selection in Indian Chief and Jarvis altered N uptake and allocation (Comstock et al., 1949; Moll and Stuber, 1971). Jarvis increased grain N via post-anthesis uptake, while Indian Chief retained more N in stover (Moll et al., 1994). Comparing Cycle 0 vs. Cycle 14 highlights regions with large allele-frequency shifts, though drift is also substantial in small populations (Gerke et al., 2015).

These populations differ in germplasm, in the N-related phenotype measured, and in how genetic signals are detected, so the same genes are not expected to appear in all of them. Each analysis also yields hundreds to thousands of candidate genes, many of which overlap by chance. If N metabolism shapes all three, however, different genes in the same metabolic pathways could be implicated. We therefore compared the analyses at the pathway level, asking whether any pathway contained candidates from more analyses than expected by chance.

We combined grain amino acid profiling and GWAS in the Goodman–Buckler panel, soil-N environmental GWAS in landraces, genomic population scans of Indian Chief and Jarvis, and reanalyzed the B64 × CML14 biparental population developed for grain asparagine variation for aspartate and proline. We analyzed free and protein-bound amino acid pools, focusing on asparagine, aspartate, glutamate, glutamine, and proline because of their roles in N transport, C–N metabolism, and storage. Although candidate regions overlapped no more than expected by chance, each analysis implicated a different enzyme of the aspartate-to-homoserine pathway, the only pathway that remained significant after correction. Beyond this pathway, individual candidates pointed to the aspartate–glutamate junction, where amino acid N connects with C–N recycling, and to amino acid transport and protein turnover, with different functions highlighted in each population. Together, these results suggest that aspartate-family metabolism is a shared source of variation in maize N metabolism.

Results

Kernel free amino acid composition in the Goodman–Buckler panel is dominated by asparagine and proline

Amino acids are the main form in which nitrogen is transported and stored during remobilization (Masclaux-Daubresse et al., 2010b; Santiago and Tegeder, 2020). During senescence and grain filling, proteins in source leaves are degraded by autophagy and vacuolar proteases (Potter and Feller, 2026), and the released nitrogen is moved to the grain largely as the amides Gln and Asn (Teixeira et al., 2026), loaded into the phloem by transporters such as AAPs and UmamiTs (Santiago and Tegeder, 2020). Most of the seed nitrogen can come from these remobilized sources (Masclaux-Daubresse et al., 2010b). The free amino acid pool of the grain is therefore the soluble, metabolically active fraction of grain nitrogen through which delivered nitrogen passes before it is stored in protein, and its composition can reveal variation in how nitrogen is supplied and used that total protein alone does not show.

FAA composition was evaluated across 280 of the 282 lines in the Goodman–Buckler panel grown in four replicates (two years × two blocks) at the Genetics Farm near Columbia, Missouri (Supplemental Tables S1–S2). To assess relative abundance, amino acids were ranked within each genotype based on average values across replicates. Asn and Pro consistently showed the highest relative abundance (Supplemental Figure S1A–B). Asn ranked as the most abundant amino acid in 146 genotypes (52%), while proline ranked first in 118 genotypes (42%). Notably, Asn and Pro comprised the top two amino acids in 179 of 280 genotypes (64%), regardless of order (Supplemental Figure S1B). This rank dominance reflected large differences in abundance. Asn was about 144 times as abundant as Cys, with Pro next, and Met and Cys at the low end (Supplemental Figure S1C). How abundant an amino acid was did not predict how much it varied among lines (Supplemental Table S3). Met varied the most (coefficient of variation, CV, 180%) despite its low abundance, followed by Gln (CV 130%). In contrast, Glu, Trp, and Gly varied the least (CV 44–54%). Asn and Pro were moderately variable (CV 70% and 63%, close to the median of 72% across the 20 FAAs), but their absolute differences among lines were still large.

Differences among lines were consistent across years. For the 251 lines grown in both years, values were correlated between years for all free amino acids (Spearman ρ = 0.33–0.75, median 0.56; e.g., Asn 0.65, Pro 0.67; Supplemental Table S3). Median broad-sense heritability (H^2) was 0.75 for the 20 FAAs (0.47–0.88), compared to 0.60 (0.49–0.68) for the 15 PBAAs measured by Shrestha et al. (2022), estimated with the same model, indicating that most variation among line means reflects genotype rather than field noise. The PBAA set is smaller because acid hydrolysis converts Asn to Asp and Gln to Glu (reported together as Asx and Glx) and destroys Trp and Cys, and Gly was not reported. The most heritable FAAs were Lys (0.88), Asn (0.83), Asp (0.81), and Pro (0.80). The most variable (Met, Gln) were also measured reliably (0.78, 0.69). Median genomic heritability (h^2) from the marker-based relationship matrix was 0.48 for FAAs and 0.34 for PBAAs, suggesting substantial additive variation suitable for association mapping (Supplemental Figure S2; Supplemental Table S4, including ratio traits).

To quantify the free pool's contribution to grain amino acid N, we compared genotype level FAAs and PBAAs in matched taxa (n = 279; Supplemental Figure S3, Supplemental Table S5) (Shrestha et al., 2022). The protein-bound pool dominated, exceeding the free pool 47-fold for total amino acids and 42-fold for amino acid-derived N (each amino acid weighted by its number of N atoms; paired Wilcoxon tests, all $P < 10^{-40}$), so FAAs contributed only 2.4% of grain amino acid-derived N. Pool composition, however, differed. To compare the two pools on the same basis, we combined free Asp+Asn and Glu+Gln to match Asx and Glx and excluded Gly, Trp, and Cys (8.4% of the free pool) from FAA totals. On this basis, Pro comprised 23.1% of FAAs versus 10.6% of PBAAs, and Asx+Glx+Arg+Lys+His comprised 56.0% versus 26.5%. Considering only the N-rich residues that are

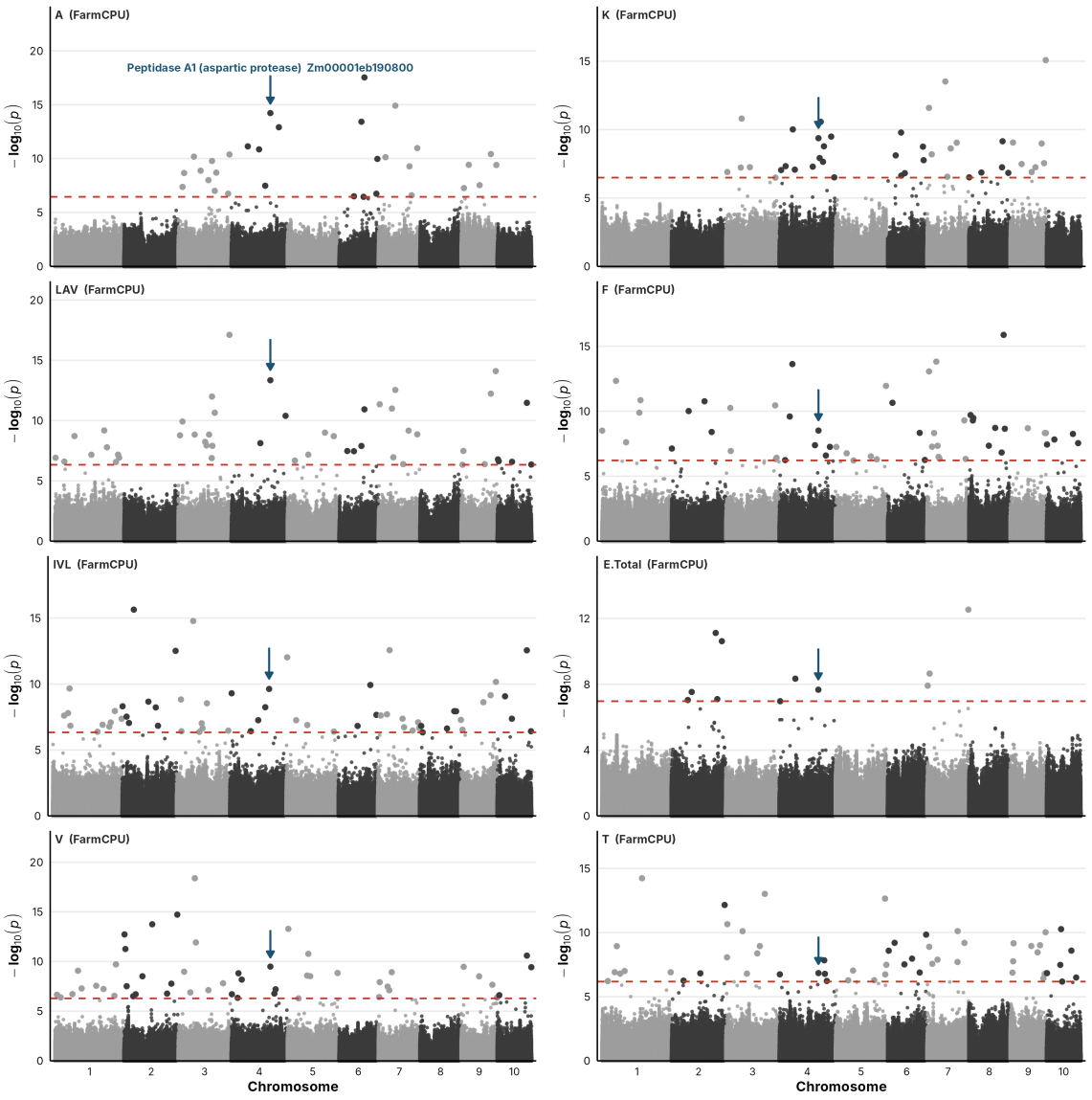


Figure 1 Manhattan plots associated with eight amino acid traits link to an aspartic protease gene A chromosome 4 locus associated with alanine (*A*), *LAV*, *IVL*, valine (*V*), lysine (*K*), phenylalanine (*F*), *E.Total*, and threonine (*T*) lies adjacent to a peptidase A1 (aspartic protease) gene. FarmCPU Manhattan plots show the significant traits at chr4:174.55–174.57 Mb. Arrows indicate the top SNP within *Zm00001eb190800*. Red dashed lines mark the 5% Benjamini–Hochberg FDR thresholds.

resolved in the free pool, Asn+Gln+Arg+Lys+His accounted for 35.7% of FAAs. Genotypic variation also differed. The Pro fraction had a coefficient of variation (CV) of 44% among lines in FAAs compared to 8% in PBAAs, and the amide+basic fraction was 20% compared to 6% (Supplemental Table S5). Thus, the free pool is small but compositionally variable among lines and consistent across years, dominated by Asn and Pro, and enriched for N-rich residues, whereas the protein-bound composition varies much less. Because the differences in free amino acids among lines were large and reproducible across years, the free pool was well suited for association mapping.

Association mapping identifies amino acid metabolism and transport candidates for abundant, heritable free amino acids

To identify loci affecting grain amino acid composition, we defined 78 amino acid–derived phenotypes (Supplemental Tables S1–S2), including the absolute abundance of 20 individual FAAs; six biosynthetic-family composites (IMNTDK, EHPRQ, SGC, WFY, LAV, IVL); 20 ratios to the total free pool; 25 ratios to the family composite; four ratios between amino acids (N.D, N.E, N.Q, Q.E); and three whole-pool measures (total FAA, total PBAA (Shrestha et al., 2022), and estimated nitrogen).

We ran GWAS using three models (MLM, BLINK, and FarmCPU). At a BH FDR of 5%, we found 2,461 significant SNPs across 71 phenotypes, forming 2,152 loci and yielding 3,693 candidate genes (Supplemental Tables S6–S7). Most candidate genes were specific to

one phenotype (3,002), 691 were shared by two or more phenotypes, and only 193 by more than two. The number of candidate genes per phenotype was similar across biosynthetic families (71, 69, and 67 for Asp-, Glu-, and Ser-family phenotypes), so families with more phenotypes simply had more candidate genes. The loci associated with the most phenotypes were not near amino acid biosynthesis genes. Three loci were each associated with nine phenotypes. The locus on chromosome 4 (212.55 Mb) had no gene within 25 kb, and the locus on chromosome 7 (40.80 Mb) was near a K homology domain protein gene (*Zm00001eb306240*). The locus on chromosome 9 (133.93 Mb) contained four genes, including an ER membrane protein complex subunit 4 gene (*Zm00001eb394360*) and a rhomboid-like protein gene (*Zm00001eb394380*), but its nine phenotypes were explained by only three groups of SNPs (Supplemental Figure S4), because many phenotypes are ratios or composites of the same amino acids. A fourth locus on chromosome 4 at 174.57 Mb was associated with eight phenotypes from four biosynthetic families (*A*, *V*, *K*, *F*, *T*, *LAV*, *IVL*, and *E.Total*), with significant SNPs 2.6–6.1 kb upstream and 5.9 kb downstream of a peptidase A1 (aspartic protease) gene (*Zm00001eb190800*; Figure 1).

Furthermore, we investigated individual FAAs (Table 1). Asn, the most abundant ($H^2 = 0.83$, $h_g^2 = 0.68$), had the most loci (356 across abundance and five ratios) and candidates, including two asparagine synthetase domain genes (*Zm00001eb036060*, *Zm00001eb036070*; *N.E*), a glutamine amidotransferase type 2 (*Zm00001eb329710*; *N.E*, $P = 6.1 \times 10^{-10}$), homoserine dehydrogenase (*Zm00001eb405410*; *N.IMNTDK*, SNP within gene), and lysine-branch enzymes dihydrodipicolinate synthase/reductase (*Zm00001eb389290*, *Zm00001eb176170*). An AVT6C transporter (*Zm00001eb368680*) was significant for Asn ($P = 1.3 \times 10^{-8}$) and the Asp-family composite ($P = 2.5 \times 10^{-11}$) in both BLINK and FarmCPU. For lysine, the most heritable FAA ($H^2 = 0.88$, $h_g^2 = 0.73$), a SNP for lysine as a fraction of the total free pool (*K.Total*; $P = 2.5 \times 10^{-11}$; significant in BLINK and FarmCPU) lies in *Zm00001eb277160*, an NRT1/PTR family 4.6 transporter gene. Another SNP for lysine relative to aspartate-family amino acids (*K.IMNTDK*; $P = 2.8 \times 10^{-11}$) is 15 kb from *Zm00001eb360480*, which encodes NADH-dependent glutamate synthase.

Pro, the second most abundant FAA, was nearly as heritable as Asn ($H^2 = 0.80$), but it had a lower genomic heritability ($h_g^2 = 0.40$), suggesting that the markers capture less of its variation among lines. It also had fewer loci (59). However, its ratio traits had no significant associations. One proline SNP lies within a class I glutamine amidotransferase gene (*Zm00001eb217220*; $P = 2.2 \times 10^{-9}$). Glutamine, the second most variable FAA (CV 130%) with high genomic heritability (0.85), showed the strongest association among the amino acids in Table 1 ($P = 4.9 \times 10^{-22}$, *Q.E*). A SNP for free glutamine (*Q*; $P = 1.2 \times 10^{-11}$) lies within a WAT1-related (UmamiT) gene on chromosome 9 (*Zm00001eb379220*). A SNP for the Gln/Glu ratio (*Q.E*; $P = 8.4 \times 10^{-8}$) lies within the amino acid/auxin permease gene *aaap40* (*Zm00001eb261470*) on chromosome 6, in a 240-kb region that also contains the lysine histidine transporters *LHT1* (*Zm00001eb261480*) and *LHT2* (*Zm00001eb261440*) and three further amino acid permeases (*aaap41*, *aaap43*, *aaap44*; *Zm00001eb261490–Zm00001eb261510*). Methionine, the most variable FAA (CV 180%), had 104 loci but no trait-matched candidate gene. Its high CV likely reflects a strongly skewed distribution across lines, where the highest line had about 200 times as much Met as the lowest (Supplemental Table S3). Thus, genes involved in Asn/Asp-family metabolism and amino acid transport linked to their respective amino acids, whereas loci common to multiple phenotypes suggested roles in protein turnover rather than in single biosynthetic reactions.

To test whether the free Asp and Pro variation seen in the panel also segregates in a biparental cross, we mapped both traits in a B64 × CML14 $F_{2:3}$ population developed to map kernel free Asn, in which Asn is controlled mainly by a single QTL at *ASN3* on chromosome 1 (Fitzsimmons et al., 2026). Because B64 and CML14 were chosen for Asn rather than Asp or Pro, this analysis was exploratory. In the Goodman–Buckler panel, CML14 was among the lowest lines for free Asp, and B64 was near the median, with both being below the median for Pro (Supplemental Figure S5A,C). We identified a single QTL for each trait at the same position on chromosome 3 (peak at 35.99 Mb), with LOD 6.12 for Asp (threshold 3.70, 1,000 permutations) and 4.55 for Pro (threshold 3.63), explaining 10% and 7% of phenotypic variance, respectively (Supplemental Figure S5B,D). This shared QTL was distinct from the Asn QTL, indicating that free Asp and Pro share genetic control that is separable from Asn in this cross. The 1.5-LOD support intervals for this shared peak were similar in size and location, spanning 14.25 and 15.27 cM for Pro and Asp, respectively, and contained 1,550 and 1,578 genes (Supplemental Table S8). The interval spans the low-recombination region around the chromosome 3 centromere, so it cannot resolve a causal gene. However, within 1.5 cM of the peak, it includes the tandem pair of CC-type glutaredoxin genes (*Zm00001eb131830*, *Zm00001eb131840*) and the LHT-like amino acid transporter (*Zm00001eb132750*), all of which also lie in highly differentiated Jarvis windows (see below). Of these, only the LHT-like gene is consistently expressed in developing kernels (Supplemental Table S8).

Soil nitrogen environment GWAS of maize landraces points to candidate genes in aspartate-family and amide nitrogen metabolism

In the Goodman–Buckler panel, the free amino acid pool of the grain was dominated by Asn and Pro, which, along with Lys and Asp, were also among the most heritable free amino acids. Association mapping identified 3,693 candidate genes, including genes involved in Asn and Asp-family metabolism and amino acid transport. Because amino acids are the main transport form of nitrogen in the plant, their metabolism may also respond to nitrogen availability in the soil. However, these inbred lines represent standing variation in temperate germplasm grown under managed fertility. We therefore turned to maize landraces collected across a soil nitrogen gradient (Figure 2B) and asked whether genetic variation associated with soil N implicates similar nitrogen-metabolism functions. Because soil N reflects the historical collection environment rather than a grain phenotype, we did not equate soil N adaptation with grain amino acid composition.

We associated the georeferenced maize landrace panel (Romero Navarro et al., 2017) with soil N at each collection site (Poggio et al., 2021), using 0–5 cm as the primary phenotype and 5–15 cm for validation (highly correlated; $r = 0.84$ across 3,107 accessions; Figure 2A–B). Soil N is a property of the collection site, so accessions from the same or nearby sites share both soil N values (467 distinct values among 3,107 accessions) and ancestry. As a result, genomic heritability was ∼1.00 and remained 0.98 after removing 50 genotype PCs (permuted $h^2 \leq 0.05$; Figure 2C), reflecting shared ancestry and location

Table 1 Association mapping of the focal FAAs in the Goodman–Buckler panel. Median, coefficient of variation (CV), line-mean broad-sense heritability (H^2) and genomic heritability (h_g^2) are from Supplemental Table S3. Loci: loci (Supplemental Table S7) significant at 5% Benjamini–Hochberg FDR for at least one phenotype reporting on the amino acid; in parentheses, loci for the absolute trait alone. Best P: smallest P among those phenotypes. Trait-matched candidate genes: genes within 25 kb of a significant SNP with their putative function.

FAA	Median	CV (%)	H^2	h_g^2	Loci	Best P	Trait-matched candidate genes (putative function)
Asparagine (N)	3.783	70	0.83	0.68	356 (73)	2.8×10^{-20}	*Zm00001eb368680* (amino acid transporter AVT6C); *Zm00001eb329710* (glutamine amidotransferase type 2); *Zm00001eb405410* (homoserine dehydrogenase); *Zm00001eb036060* (asparagine synthetase domain); *Zm00001eb036070* (asparagine synthetase domain); *Zm00001eb389290* (dihydrodipicolinate synthase); *Zm00001eb176170* (dihydrodipicolinate reductase)
Proline (P)	3.202	63	0.80	0.40	59 (59)	2.7×10^{-15}	*Zm00001eb217220* (glutamine amidotransferase)
Lysine (K)	0.272	90	0.88	0.73	116 (54)	8.3×10^{-16}	*Zm00001eb277160* (NRT1/PTR family 4.6); *Zm00001eb360480* (glutamate synthase (NADH)); *Zm00001eb352400* (WAT1-related (UmamiT))
Aspartate (D)	1.178	76	0.81	0.84	91 (20)	3.3×10^{-16}	*Zm00001eb207630* (WAT1-related (UmamiT))
Glutamine (Q)	0.371	130	0.69	0.85	165 (65)	4.9×10^{-22}	*Zm00001eb379220* (WAT1-related (UmamiT)); *Zm00001eb261470* (amino acid transporter); *Zm00001eb261480* (amino acid transporter); *Zm00001eb261490* (amino acid transporter)
Methionine (M)	0.047	180	0.78	0.46	104 (56)	3.2×10^{-17}	none

rather than heritable variation in the plants. We therefore tested for confounding by relatedness and geography directly. Similarly, across three GWAS (MLM, BLINK, FarmCPU) models, 234 SNPs were significant at 5% FDR (Figure 2D). BLINK detected 218, FarmCPU 30, and MLM 2. Most SNPs were detected by a single model (203 by BLINK and 16 by FarmCPU), and 15 were detected by at least two, 13 by BLINK and FarmCPU, one by MLM and BLINK, and one (S5_22338979) by all three. In total, 448 genes lie within 25 kb of 216 of these SNPs (Supplemental Table S9).

Because soil N varies across the landscape and landrace ancestry follows geography, an association with soil N can arise from shared ancestry or from other site variables that covary with soil N, rather than from soil N itself. We therefore filtered the soil N SNPs in several steps, using two tests to assess whether associations reflected soil N rather than geography or climate, and then focusing on SNPs detected by more than one model. The first asked whether the same SNPs were detected when other site variables were used as the GWAS phenotype. We ran GWAS with the same models for latitude, longitude, elevation, soil pH, CEC, and aridity and counted SNPs shared with soil N (Supplemental Figure S6; Supplemental Tables S10–S11). Only 2 of the 234 soil-N SNPs were also significant for latitude, longitude, or elevation, close to the 0.7 expected by chance ($P = 0.14$). By contrast, soil N shared many SNPs with aridity (38 compared to 0.2 expected) and soil pH (13 compared to 0.1), which covary with it. In total, 190 of the 234 SNPs were not significant for any other variable. Because a SNP can follow a geographic gradient without reaching significance in another GWAS, the second test did not use a GWAS model. For every SNP in the genome, we calculated the correlation between genotype and each site variable across the 3,107 accessions and considered a soil N SNP linked to another variable if its correlation was in the top 1% of the genome for that variable (Supplemental Figure S7; Supplemental Tables S12–S13). By this measure, 13.2% of soil N GWAS SNPs were linked to a geographic variable, compared with 1.8% of all SNPs and 35.9% of SNPs selected only for their correlation with soil N ($P = 5.9 \times 10^{-15}$). The GWAS models therefore removed much, but not all, of the geographic signal. Overall, 168 of the 234 SNPs passed both tests. Because associations detected by a single model are less reliable, we focused on the 15 SNPs detected by at least two models. Of these, 14 passed the first test and 12 passed both (Table 2). The three that failed included the strongest signal (S9_151742636; $P = 2.6 \times 10^{-13}$), which was also significant for aridity and soil pH. Of the 12, five had genotypes correlated with soil N but not with other site variables. The other seven had genotypes only weakly correlated with soil N itself, so the lack of correlation with other variables was uninformative for them.

Four soil N SNPs map near genes involved in Asp-family or amide nitrogen metabolism. S5_22338979 lies 4.4 kb upstream of *Zm00001eb220040*, an amino acid kinase-domain gene annotated with aspartate kinase activity, the first step of the aspartate-derived amino acid pathway. It is the only soil-N SNP significant in all three models and is also significant at 5–15 cm. S1_16128632 lies within *Zm00001eb005710*, an ω-amidase/NIT2-like CN-hydrolase (Zhang and Marsolais, 2014). Both SNPs are rare (minor allele frequencies of 0.030 and 0.016), so their associations rest on few accessions. Because both SNPs are rare, we examined where the minor-allele carriers were collected (Supplemental Figure S8). Carriers of the ω-amidase allele came from 56 collection sites, and carriers of the aspartate kinase allele came from 115 sites, each spanning 11 countries, so neither association rests on a single site or local population. Both alleles were concentrated in Mexico and Central America (87% and 93% of carriers, compared with 74% of the panel) and were nearly absent from South America (5 of 667 accessions for each allele). Carriers came from soils with more N than non-carriers (median 414 compared with 251 for ω-amidase and 356 compared with 248 for aspartate kinase), and both heterozygous and homozygous carriers showed this shift (Kruskal–Wallis $P = 4.4 \times 10^{-9}$ and 3.3×10^{-10}). Because soil N also differs between regions, we compared carriers with non-carriers within regions. Within Mexico, ω-amidase carriers came from soils with nearly twice the N of non-carriers (median 424 compared with 233, 28 carriers, Wilcoxon $P = 1.0 \times 10^{-4}$), whereas the difference within Central America was smaller (451 compared with 404, 25 carriers, $P = 0.32$). For aspartate kinase, the differences within Mexico (268 compared with 232, 75 carriers, $P = 6.4 \times 10^{-4}$) and within Central America (445 compared with 403, 39 carriers, $P = 0.06$) were smaller, indicating that part of its overall association reflects the higher soil N of Central America, where the allele is more common, although carriers still came from N-richer soils within Mexico. Thus, the minor alleles at both loci are more frequent at N-rich collection sites, most clearly for ω-amidase within Mexico, although the regional concentration of carriers means that shared ancestry cannot be excluded. S8_138640842, the strongest of the 12 SNPs that passed both tests ($P = 3.5 \times 10^{-11}$), lies within *Zm00001eb356170* (methionine S-methyltransferase). However, it was supported only by BLINK and FarmCPU and was inconclusive in the correlation test. At 5–15 cm, S8_133179205 lies 0.1 kb from *Zm00001eb354640* (S-adenosylmethionine synthase) and was not

Table 2 Soil-N SNPs detected by at least two GWAS models that passed both environmental specificity tests. SNPs were FDR-significant (5%) for soil N at 0–5 cm in at least two of MLM (M), BLINK (B) and FarmCPU (F). They were not significant in the GWAS of latitude, longitude, elevation, soil pH, CEC or aridity (Supplemental Table S11), and their genotype was not in the top 1% of the genome for correlation with any of those variables (Supplemental Table S12). P, lowest P across models. 5–15 cm, also FDR-significant for soil N at 5–15 cm. Corr., genotype–environment call, either not linked (genotype in the top 5% for correlation with soil N and not in the top 1% for any other variable) or inconclusive (genotype outside the top 5% for correlation with soil N, so the association is detected only by the multi-locus models). Gene (distance), gene within 25 kb of the SNP and its distance, or within if the SNP lies in the gene.

SNP	Chr	Position	Models	MAF	P	5–15 cm	Corr.	Gene (distance)	Annotation
S8_138640842	8	138,640,842	BF	0.254	3.5×10^{-11}	no	inconclusive	*Zm00001eb356170* (within)	Methionine S-methyltransferase
S1_87643408	1	87,643,408	BF	0.356	2.6×10^{-10}	no	inconclusive	*Zm00001eb022740* (within)	Syndetin
S2_191754648	2	191,754,648	BF	0.074	6.2×10^{-10}	no	inconclusive	*Zm00001eb100890* (within)	Uncharacterized protein
S5_22338979	5	22,338,979	MBF	0.030	1.9×10^{-8}	yes	not linked	*Zm00001eb220040* (4.4 kb)	Amino acid kinase domain
S8_153826994	8	153,826,994	BF	0.020	5.1×10^{-8}	no	inconclusive	*Zm00001eb359990* (within)	Homeobox-DDT protein RLT1
S3_54622531	3	54,622,531	BF	0.050	6.5×10^{-8}	yes	not linked	*Zm00001eb130210* (17.9 kb)	RING/FYVE/PHD zinc finger (TTL1-like)
S5_22338938	5	22,338,938	MB	0.031	6.5×10^{-8}	no	not linked	*Zm00001eb220040* (4.3 kb)	Amino acid kinase domain
S1_16128632	1	16,128,632	BF	0.016	2.9×10^{-7}	no	not linked	*Zm00001eb005710* (within)	ω-amidase/NIT2-like CN-hydrolase
S5_53136163	5	53,136,163	BF	0.221	5.7×10^{-7}	yes	inconclusive	*Zm00001eb226000* (within)	Protein kinase
S6_37926580	6	37,926,580	BF	0.179	7.4×10^{-7}	no	inconclusive	*Zm00001eb265710* (within)	HB transcription factor
S1_5543355	1	5,543,355	BF	0.090	1.2×10^{-6}	yes	inconclusive	*Zm00001eb001920* (within)	Transcription repressor
S2_117202525	2	117,202,525	BF	0.071	2.1×10^{-6}	no	not linked	–	No gene within 25 kb

linked to other site variables. The same gene was a candidate in the grain amino acid GWAS (*I.IVL*; $P = 1.0 \times 10^{-8}$). Thus, the soil-N GWAS in landraces, like the grain amino acid GWAS, points to Asp-family and amide nitrogen metabolism.

Free amino acid profiles of Indian Chief and Jarvis show population-specific remodeling of the soluble amino acid pool

The Goodman–Buckler panel captured standing variation in grain amino acids, and landraces captured variation tied to natural soil N gradients, but neither shows how the grain amino acid pool responds to yield selection. The Indian Chief and Jarvis maize populations address this question. Both were base populations in a 14-cycle reciprocal recurrent-selection experiment for yield, and selection shifted N allocation differently in seeds and stover (Moll et al., 1994). Under nonlimiting N, Jarvis Cycle 14 increased grain N mainly via greater grain weight, driven largely by higher post-anthesis soil N uptake, with little change in apparent stover-to-grain N remobilization. Indian Chief Cycle 14 maintained more post-anthesis stover biomass, lost less stover N after anthesis, and did not significantly increase grain N (Moll et al., 1994). Thus, grain N gains can arise from improved soil N recovery or altered internal N allocation during grain filling (Mueller et al., 2019). Because amino acids deliver N to the kernel, we asked whether yield selection also remodeled the grain free amino acid pool, and whether populations with different N responses changed differently. We therefore treat the Indian Chief and Jarvis amino acid profiles as evidence of changes in the grain amino acid pool, not of changes in N allocation or remobilization.

We compared total FAA, the fractions of Pro, Asn, Asp, Gln, and Glu in the free pool, the N-rich FAAs (Asn + Gln + Arg + Lys + His), and FAA-derived N between Cycle 0 and Cycle 14 within each population, correcting for these eight tests (Figure 3; Supplemental Table S14). We also tested all 20 FAA fractions and the protein-bound pool separately (Supplemental Tables S15–S17). Indian Chief showed significant changes in total FAA and composition. Total FAA per mg increased from 1.40 (Cycle 0) to 2.46 (Cycle 14) (Wilcoxon $P = 0.0043$; BH $P = 0.017$; Figure 3A), with no overlap between cycles, and FAA-derived N increased accordingly (1.75 to 2.99 per mg; BH $P = 0.017$). Pro increased from 0.320 to 0.519 of total FAA ($P = 0.016$; BH $P = 0.025$), whereas Asp decreased from 0.115 to 0.075 ($P = 0.016$; BH $P = 0.025$), and Glu from 0.140 to 0.062 ($P = 0.0068$; BH $P = 0.018$). Asn and Gln fractions were unchanged (Figure 3B). When all 20 FAA fractions were tested together, Pro, Asp, Glu, Ile, and Val differed nominally ($P \leq 0.016$), but none remained significant after correction across the 20 tests (BH $P = 0.063$; Supplemental Table S15).

Jarvis showed one nominal change, but the test was underpowered with only three Cycle 0 samples. The Asp fraction increased from 0.075 (Cycle 0) to 0.137 (Cycle 14; $P = 0.019$), but it was not significant after correction (BH-adjusted $P = 0.15$). Total FAA per mg, the Pro and Glu fractions, the N-rich FAA pool, and FAA-derived N did not differ between cycles (Supplemental Table S14). The lack of corrected differences in Jarvis may reflect limited sampling, a stable FAA pool, or both. A stable pool is also consistent with Moll et al. (1994), in which the greater grain N of Jarvis Cycle 14 came mainly from greater grain weight and post-anthesis N uptake rather than from increased remobilization from the stover.

To test whether changes in the free pool extended to total grain amino acid N, we converted FAA and PBAA abundances to amino acid derived N per mg. Protein-bound composition did not change across cycles in either population (Supplemental Figure S9; Supplemental Tables S15–S16). In Indian Chief, FAA-derived N increased from 1.75 to 2.99 per mg, but PBAA-derived N was more than 100-fold larger (332 and 364 per mg) and did not change, nor did total amino acid derived N ($P = 0.42$ for both; Supplemental Figure S10; Supplemental Table S17). Because FAAs carry less than 1% of grain amino acid N, the larger free pool in Cycle 14 did not measurably change grain amino acid N, consistent with unchanged grain N in Indian Chief (Moll et al., 1994). Jarvis showed no change in any amino acid derived N pool. Thus, after 14 cycles of selection, Indian Chief had a greater total FAA and a composition that shifted from Asp/Glu toward Pro, whereas Jarvis showed no significant change. The protein-bound pool, which contains most grain amino acid N, was unchanged in both populations. These population-specific responses are consistent with reported differences in N allocation (Moll et al., 1994).

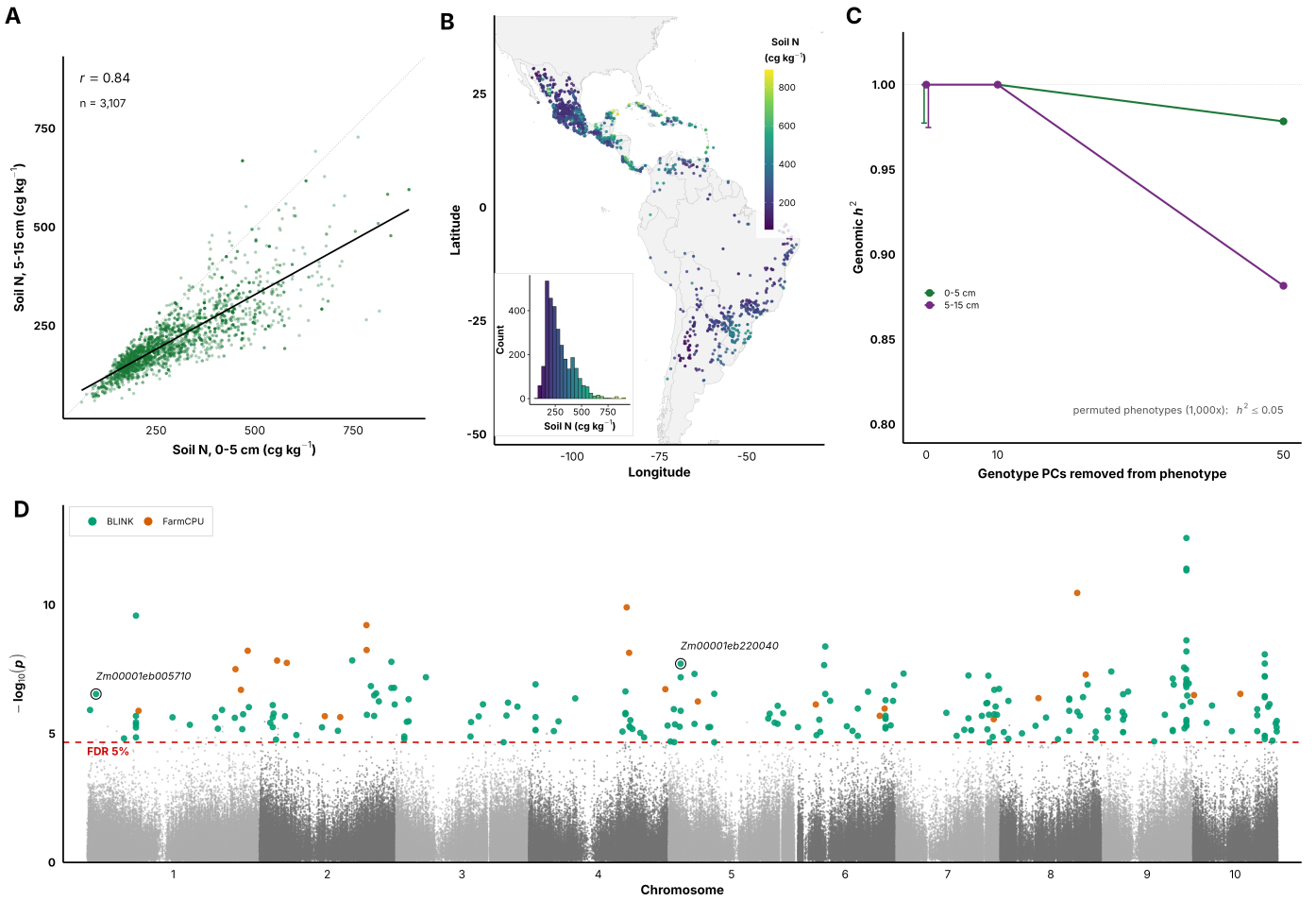


Figure 2 Soil nitrogen at the collection site of 3,107 maize landraces. (A) Soil N at 0–5 cm vs. 5–15 cm ($r = 0.84$); solid line, least-squares fit; dotted line, $y = x$. **(B)** Collection sites colored by soil N at 0–5 cm; inset, distribution of soil N across accessions. **(C)** Genomic heritability (h^2) of soil N at both depths after removing 0, 10 and 50 genotype principal components from the phenotype; error bars, 95% CI at 0 PCs; permuted phenotypes (1,000×) gave $h^2 \leq 0.05$. **(D)** Manhattan plots for soil N (0–5 cm); colored points are SNPs significant at 5% Benjamini–Hochberg FDR in at least one model (MLM, BLINK, FarmCPU; 3 PCs), colored by the model with the lowest P; the SNPs near the aspartate kinase isozyme (*Zm00001eb220040*) and the ω-amidase (*Zm00001eb005710*) are ringed and labeled. The dashed red line marks the 5% FDR threshold. Candidate genes named in the text are labeled.

Skim sequencing reveals that Indian Chief and Jarvis diverged during reciprocal recurrent selection

We skim sequenced 379 individuals from the Indian Chief and Jarvis populations at Cycle 0 and Cycle 14 (Indian Chief 96 and 92; Jarvis 96 and 95). The mean sequencing depth was 0.73× per individual and was higher at Cycle 14 than at Cycle 0 in both populations (Indian Chief 0.61× compared with 0.88×; Jarvis 0.68× compared with 0.76×; Supplemental Figure S11). At this depth, about half of the genotypes in each individual could not be called, and because the number of calls depends on depth, hard-called genotypes would differ between cycles for technical rather than biological reasons. We therefore used genotype likelihoods throughout, estimating population structure from 512,667 filtered SNPs (Supplemental Figure S12) and F_{ST} directly from the aligned reads with ANGSD (Korneliussen et al., 2014).

Unsupervised ancestry estimation ($K = 2$–5) on sites thinned to one per 10 kb separated the four population cycle groups (Figure 4A). Likelihood increased strongly to $K = 4$, where runs converged and each group formed a distinct cluster. However, at $K = 5$, runs disagreed, and likelihood gains were small. At $K = 2$–3, Indian Chief Cycle 0 individuals were inferred as a mixture of the Jarvis Cycle 0 and Indian Chief Cycle 14 clusters, placing the ancestral Indian Chief population between Jarvis and its selected descendant. Genotype-likelihood PCA showed the same pattern (Figure 4B). PC1 (14.3%) separated Indian Chief Cycle 14 from Jarvis Cycle 14, with both Cycle 0 groups near the center, and PC2 (6.7%) separated Cycle 14 from Cycle 0. Overall, the populations, initially less differentiated at Cycle 0, diverged in opposite directions over 14 cycles.

Genetic differentiation confirmed the divergence between the populations. F_{ST} between Indian Chief and Jarvis nearly doubled from 0.111 at Cycle 0 to 0.211 at Cycle 14, increasing across all ten chromosomes (Figure 4C). Because Cycle 14 had deeper sequencing, which should improve allele-frequency estimates and reduce F_{ST}, the increase is unlikely to be a depth artifact. This divergence is expected under reciprocal recurrent selection, which favors complementary rather than shared alleles (Comstock et al., 1949) and is reinforced by drift in small, separately maintained populations. Thus, Indian

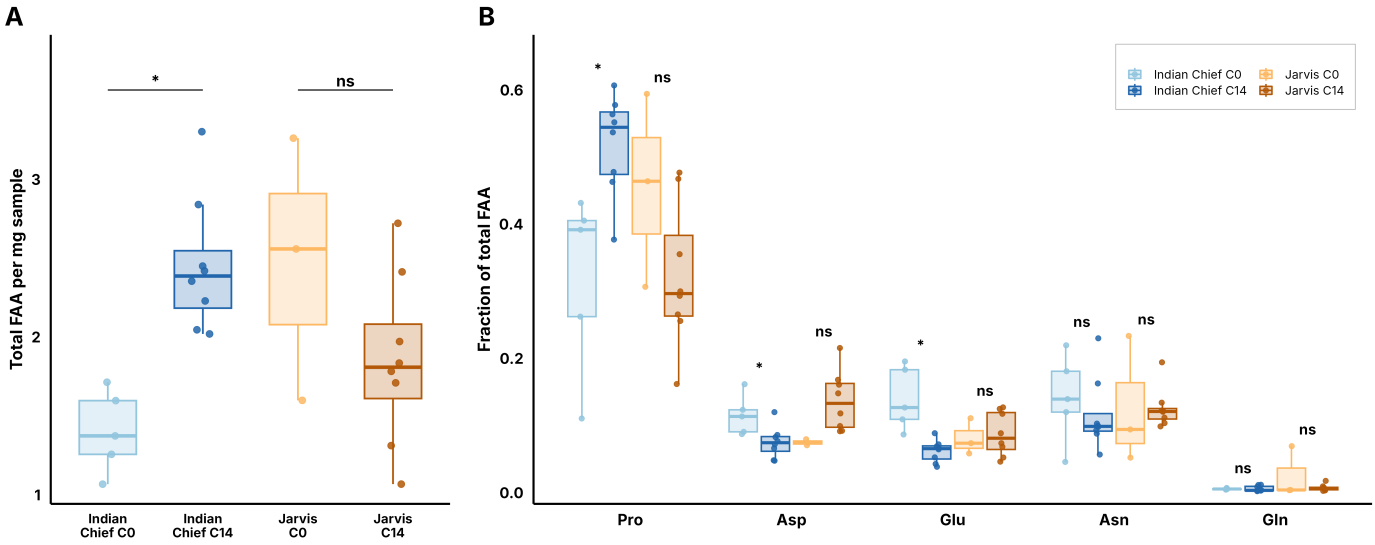


Figure 3 Free amino acid (FAA) pool and composition in Indian Chief and Jarvis before (C0) and after (C14) 14 cycles of reciprocal recurrent selection. **(A)** Total FAA per mg sample. **(B)** Fractions of total FAA for Pro, Asp, Glu, Asn and Gln. Points are biological samples, each the mean of three technical replicates; boxes show the median and interquartile range. Within each population, C0 and C14 were compared with two-sided Wilcoxon rank-sum tests, and P values were adjusted with the BH method; * adjusted $P < 0.05$; ns, not significant.

Chief and Jarvis became more differentiated from each other over 14 cycles of reciprocal recurrent selection.

F_{ST} scans identify population-specific candidate regions in Indian Chief and Jarvis

Before scanning for regions of high differentiation between cycles, we asked how much of the allele frequency change could be explained by genetic drift. In the reciprocal recurrent selection program, each cycle was formed by intermating the self-pollinated progenies of the 20 best plants per population (Moll and Stuber, 1971), which implies an effective population size (N_e) of at most 20 per cycle. Genome-wide mean F_{ST} between Cycle 0 and Cycle 14 was 0.095 in Indian Chief and 0.114 in Jarvis (Figure 5), values reproduced by neutral Wright–Fisher simulations at $N_e \approx 30$ and $N_e \approx 26$ per cycle, respectively (Supplemental Figure S13). Although slightly above the design-based expectation, these estimates indicate that most genome-wide change between cycles is consistent with drift in small populations. A similar conclusion was reached for the Iowa Stiff Stalk Synthetic (BSSS) and Iowa Corn Borer Synthetic No. 1 (BSCB1), two maize populations under reciprocal recurrent selection for hybrid yield, where linked drift simulations from founder haplotypes explained most of the loss of diversity, and few regions deviated from drift or were shared between populations (Gerke et al., 2015). At this N_e, drift moves long linked regions together, so individual outlier regions cannot be formally distinguished from drift. We therefore treat the scans below as exploratory.

We scanned each population for the regions with the largest allele-frequency change between Cycle 0 and Cycle 14 using F_{ST} summarized in windows (2,162 windows in Indian Chief and 2,151 in Jarvis; median length 500 kb). We retained the top 5% of windows ($F_{ST} \geq 0.157$ in Indian Chief and ≥ 0.178 in Jarvis; 109 and 108 windows), which contained 2,001 and 1,179 genes, respectively (Figure 5; Supplemental Tables S18–S21). Candidate regions were largely population-specific. Only five regions (2.5 Mb; 73 genes) were in the top 5% in both populations, no more than expected by chance ($P = 0.80$; Figure 5), indicating that the two populations show no evidence of shared targets of differentiation. Likewise, 192 of the 2,152 Goodman–Buckler amino acid GWAS loci (8.9%) fell in a top-5% window of either population, the same proportion as the genome covered by these windows (8.9%; $P = 0.52$). Thus, the F_{ST} scans and the amino acid GWAS do not converge on the same genomic regions beyond chance.

Each candidate set contains more than 1,000 genes, so genes from almost any metabolic pathway are expected to be present by chance. We therefore focused on genes in the pathways highlighted by the amino acid and soil N GWAS (Asp-family and Glu/Gln metabolism, amino acid and N transport, and protein turnover) and described them as candidates of interest rather than as evidence of selection. Pathway reoccurence across analyses more than expected is tested in the next section. In Indian Chief, candidate regions included genes of the Asp-family amino acid metabolism, *Zm00001eb214360* (aspartate-semialdehyde dehydrogenase) and *Zm00001eb294790* (threonine synthase), and genes of Glu/Gln interconversion, including two glutamate dehydrogenases (*Zm00001eb060520*, *Zm00001eb162300*), a glutamine synthetase domain-containing protein (*Zm00001eb162900*), and a glutamine amidotransferase domain-containing protein (*Zm00001eb217220*), which also contains a significant SNP for proline in the Goodman–Buckler GWAS (Table 1). Candidate regions also contained genes involved in N transport and protein turnover, including the senescence-specific cysteine protease *SAG39* (*Zm00001eb214780*) and an ammonium transporter (*Zm00001eb214890*), which lie in the same window on chromosome 5; an NRT1/PTR-family transporter (*Zm00001eb287950*) and the autophagy-related gene *ATG18g* (*Zm00001eb287690*), which share a window on chromosome 6, as well as two further autophagy-related genes (*Zm00001eb352300* and *Zm00001eb372060*). These

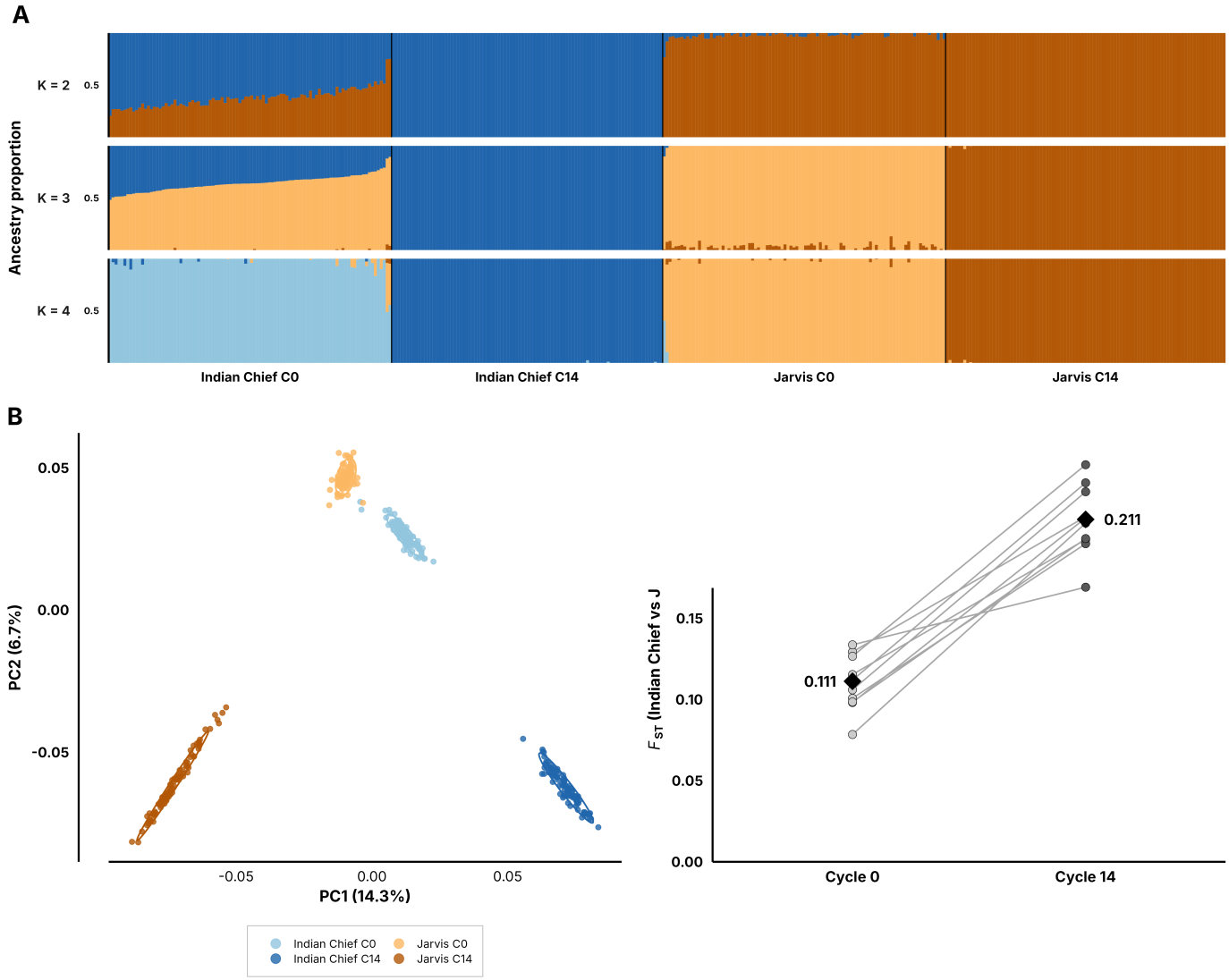


Figure 4 Indian Chief and Jarvis divergence during 14 cycles of reciprocal recurrent selection. All panels use genotype likelihoods. **(A)** Unsupervised ancestry proportions for $K = 2$–4 (thinned to one per 10 kb); each bar is one individual, and each cluster is colored by the group with the highest mean membership. $K = 4$ is the largest K at which independent runs converged. **(B)** Principal component analysis of the genotype-likelihood covariance matrix; ellipses enclose 90% of each group. Individuals with lower depth carry less information and lie closer to the center of their group. **(C)** F_{ST} between Indian Chief and Jarvis at Cycle 0 and Cycle 14, estimated from the joint site-frequency spectrum of each pair; gray lines connect the same chromosome and black diamonds give the genome-wide value.

annotations are compatible with the Indian Chief FAA phenotype, in which total FAA and the Pro fraction increased while the Asp and Glu fractions decreased from Cycle 0 to Cycle 14, but they do not establish that these genes underlie it.

In Jarvis, candidate regions included two genes of Asp metabolism, including *Zm00001eb094670*, annotated as aspartate kinase, and an aspartate aminotransferase (*Zm00001eb152450*, GOT1/THP3). These genes are of interest because the Asp fraction was the only FAA trait that changed in Jarvis, increasing nominally from Cycle 0 to Cycle 14. Jarvis candidate regions also contained two amino acid transporter genes (*Zm00001eb132750*, an amino acid transporter transmembrane domain-containing protein, and *Zm00001eb159580*, a putative GABA transporter), NPL4-like protein 1 (*Zm00001eb146560*), which acts in ubiquitin-dependent protein degradation, and two glutaredoxins (*Zm00001eb131830* and *Zm00001eb131840*). As in Indian Chief, these candidates point to amino acid metabolism, transport, and protein turnover as processes worth testing, but they require functional validation.

The aspartate-to-homoserine pathway recurs across the three populations

To test whether the three populations implicate the same metabolic pathways, we asked whether any CornCyc pathway (Hawkins et al., 2021) contained candidate genes from more than one of the four analyses (amino acid GWAS, soil N GWAS, and the Indian Chief and Jarvis F_{ST} scans) more often than expected by chance. We represented each candidate set by its genomic intervals and generated a null distribution by randomly shifting the intervals

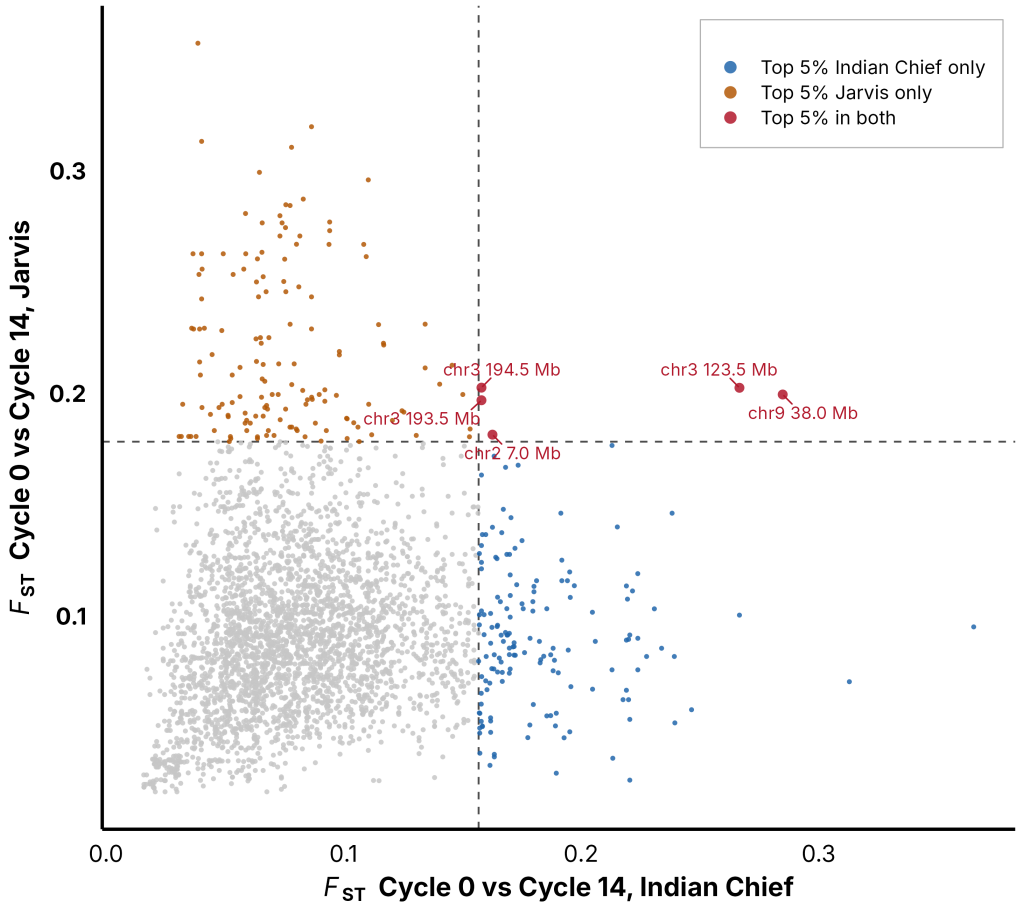


Figure 5 Candidate regions of differentiation between Cycle 0 and Cycle 14 F_{ST} between Cycle 0 and Cycle 14 in Indian Chief (x-axis) and Jarvis (y-axis), estimated from genotype likelihoods and summarized in spline-defined windows. Each point is a genomic segment bounded by the window boundaries of either population, and carries the F_{ST} of the Indian Chief and Jarvis windows that contain it. Dashed lines mark the top 5% of windows in each population ($F_{ST} \geq 0.157$ in Indian Chief and ≥ 0.178 in Jarvis). Blue and orange, top 5% in one population only; red, top 5% in both, labeled with chromosome and position.

along each chromosome, which preserves their number, size, and clustering. Because this test asks whether one pathway stands out from all others, larger candidate sets add background genes that hit many pathways by chance. Therefore, we used stricter sets than in the gene level analyses. The bonferroni significant SNPs for both GWAS and the top 3% of F_{ST} windows. Among 468 pathways, L-homoserine biosynthesis had candidates from all four analyses, had the lowest permutation-based empirical P (P = 0.00034), and was the only pathway that remained significant after correction across all pathways (corrected $P = 0.021$; Figure 6). Each analysis implicated a different enzyme, homoserine dehydrogenase (*Zm00001eb405410*) in the amino acid GWAS (Asn relative to the Asp-family amino acids, *N.IMNTDK*), an aspartate kinase isozyme (*Zm00001eb220040*) in the soil N GWAS, aspartate-semialdehyde dehydrogenase (*Zm00001eb214360*) in Indian Chief, and aspartate kinase (*Zm00001eb094670*) in Jarvis. L-lysine biosynthesis VI also had candidates from all four analyses (P = 0.0030), largely because it shares the aspartate kinase and aspartate-semialdehyde dehydrogenase steps. The other pathways with $P < 0.05$ were small terpenoid and vitamin E pathways with candidates from two or three analyses, and none remained significant after correction.

We reran the test across 12 combinations of GWAS thresholds (FDR 5%, 3%, 1%, and Bonferroni) and F_{ST} window thresholds (top 5%, 3%, 1%) (Supplemental Figure S14). The same four genes were recovered at all GWAS thresholds when using the top 5% or 3% of F_{ST} windows, and in all eight such combinations, L-homoserine biosynthesis ranked first or second among 468 pathways. Across all 12 combinations, it ranked first five times and second three times. The lower ranks occurred only with the top 1% windows (Supplemental Figure S14B). Using larger candidate sets (FDR 5% and top 5% windows), more pathways were hit by chance, and the pathway was not significant after correction (corrected P = 0.68). Thus, L-homoserine biosynthesis ranked consistently across thresholds, while corrected significance depended on candidate-set size. Although candidate regions from the four analyses overlap no more than expected by chance, each implicates a different step in aspartate-to-homoserine conversion.

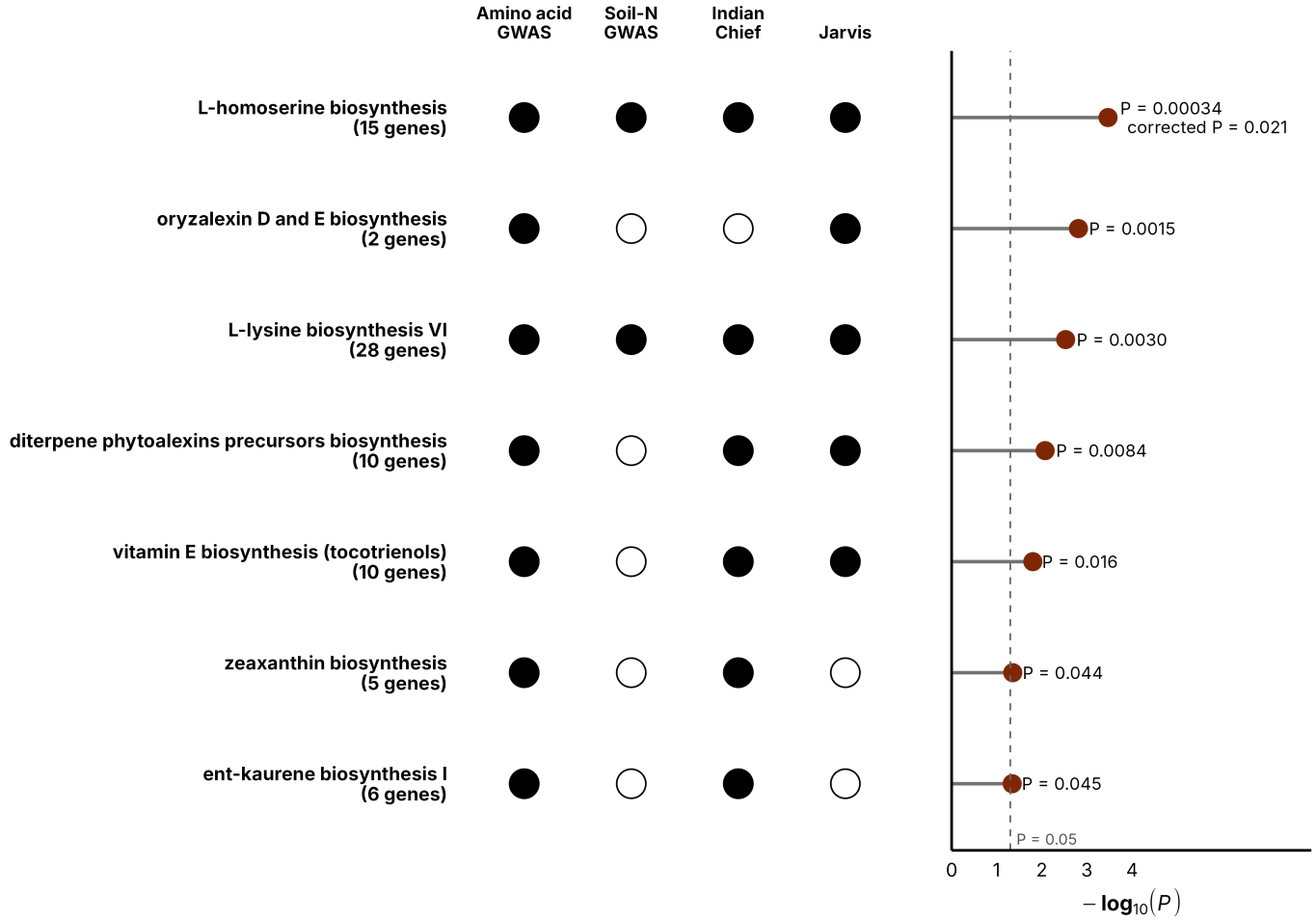


Figure 6 The L-homoserine biosynthesis pathway recurs across all four analyses. All CornCyc pathways with uncorrected permutation $P < 0.05$ for recurrence across the amino acid GWAS, soil-N GWAS and the Indian Chief and Jarvis F_{ST} scans, using Bonferroni-significant GWAS SNPs and the top 3% of F_{ST} windows. Left, filled circles mark analyses with at least one candidate gene in the pathway; numbers in brackets give the number of maize genes in the pathway. Right, pathway-specific permutation-based empirical P for the observed number of analyses hitting the pathway; dashed line, $P = 0.05$. The corrected P (minimum-P correction across all 468 pathways) is shown for the only pathway that remained significant.

Discussion

We tested whether natural variation in grain amino acids, genetic variation linked to soil N, and yield selection implicate the same N-metabolism functions in maize. In the Goodman–Buckler panel, the free amino acid pool was small, dominated by Asn and Pro, highly variable, and heritable. GWAS highlighted candidates for amino acid metabolism, transport, and protein turnover. In landraces, soil N at the collection site associated with SNPs near an aspartate kinase isozyme and an ω-amidase, surviving controls for geographic and climatic confounding. After 14 cycles of reciprocal recurrent selection for yield, Indian Chief and Jarvis differed in how they allocated N, with Jarvis increasing grain N mainly through post-anthesis N uptake and Indian Chief retaining more N in the stover (Moll et al., 1994). In our samples, Indian Chief also accumulated more free amino acids, mostly Pro, and the two populations diverged further genetically. Although most candidates were analysis-specific, each analysis implicated a different enzyme in the aspartate-to-homoserine pathway, the only pathway that remained significant after correction. Next, we analyze multi-trait loci, the aspartate-to-homoserine pathway, and the aspartate–glutamate junction, along with candidate genes involved in transport and protein turnover. These annotation-based candidates require functional validation.

Loci shared by many amino acid traits point to protein turnover and membrane protein biogenesis rather than specific biosynthetic steps

Few GWAS loci were shared across many amino acid phenotypes. Phenotype counts also overstate independent signals because composites and ratios reuse amino acids. The most interpretable case is the chromosome 4 locus at 174.57 Mb, associated with eight phenotypes (including Ala, Val, Lys, Phe, and Thr) spanning four biosynthetic families, and adjacent to *Zm00001eb190800*, encoding a peptidase A1 (pepsin-like aspartic protease) (Figure 1). Aspartic proteases are known to drive bulk protein turnover. In senescing tobacco, chloroplast CND41 degrades Rubisco, and reduced CND41 increases Rubisco retention under N limitation (Kato et al., 2004).

Rubisco breakdown is a major source of amino acids released during senescence (Feller et al., 2008), although A1 proteases also act in development and stress responses (Soares et al., 2019). A single locus affecting amino acids from four biosynthetic families is more easily explained by a change in protein degradation than by a change in one biosynthetic enzyme, which makes *Zm00001eb190800* a plausible candidate for functional testing.

The chromosome 9 locus implicates the endomembrane system. A ∼13 kb region associated with nine phenotypes (including total free amino acids) spans *Zm00001eb394360* and *Zm00001eb394380*. *Zm00001eb394360* encodes ER membrane protein complex subunit 4, which inserts transmembrane proteins into the ER (Guna et al., 2018), and *Zm00001eb394380* encodes rhomboid-like protein 19, an intramembrane serine protease annotated for Golgi-to-ER retrograde transport. Although the functions of most plant rhomboid-like proteins remain unknown (Lavell et al., 2021), both genes act in the ER–Golgi system that processes transporters and secreted proteins involved in amino acid and N movement (Brandizzi, 2018), so either could affect the overall free amino acid pool. Because the associations do not clearly separate these genes from nearby genes (including a DUF2404 membrane protein at the chromosome 4 locus), both loci should be treated as positional candidates, and kernel expression and mutant analyses are the most direct tests. Candidate annotations, trait associations, and model-specific GWAS results are available at `https://j83gk2-nirwan-tandukar.shinyapps.io/amino_gwas_app/`.

The aspartate-to-homoserine pathway links soil-N adaptation, recurrent selection, and grain amino acid variation

Asn was the most abundant free amino acid in the Goodman–Buckler panel and, because of its high N-to-C ratio, is a major N transport and storage form in plants (Lea et al., 2007). It is synthesized from Asp by asparagine synthetase using the amide N of Gln, and catabolism returns carbon to Asp or oxaloacetate (Lea et al., 2007). Asp is thus both the precursor of Asn and the entry point to the Asp-family pathway that produces Lys, Thr, Met, and Ile (Han et al., 2021; Wang et al., 2018; Viola, 2001). Pathway flux is regulated mainly by feedback inhibition of aspartate kinase by Lys and Thr, and of dihydrodipicolinate synthase by Lys (Azevedo et al., 2006; Jander and Joshi, 2009). This pathway was implicated by all four analyses. L-homoserine biosynthesis had a candidate gene in every analysis, had the lowest permutation-based empirical P of the 468 CornCyc pathways, and was the only pathway that remained significant after correction (Figure 6).

Across analyses, implicated genes spanned the Asp to homoserine pathway. Jarvis regions included aspartate kinase (*Zm00001eb094670*), the first committed step (Viola, 2001), and *ask2* expression is linked to Met accumulation in developing kernels (Duo et al., 2026). The only soil N SNP significant in all three GWAS models, which also passed both tests for confounding with geography and climate, was near another aspartate kinase (*Zm00001eb220040*). Indian Chief regions included aspartate-semialdehyde dehydrogenase (*Zm00001eb214360*), and the amino acid GWAS implicated homoserine dehydrogenase (*Zm00001eb405410*), the first committed step to Thr and Met. Further downstream, Indian Chief regions included threonine synthase (*Zm00001eb294790*) and methionine synthase (*Zm00001eb052750*), the latter also near a soil-N SNP. The amino acid GWAS also identified genes in the Lys branch, dihydrodipicolinate synthase (*Zm00001eb389290*), dihydrodipicolinate reductase (*Zm00001eb176170*), and diaminopimelate decarboxylase (*Zm00001eb406820*), and in the Asn branch, two asparagine synthetase domain genes (*Zm00001eb036060*, *Zm00001eb036070*) and the amino acid transporter AVT6C (*Zm00001eb368680*), suggesting that transport or storage, not only synthesis, may shape the Asn pool.

Together, three populations with different germplasm and study designs point to the same short pathway, with each analysis implicating a different enzyme. This pattern did not depend on a single threshold. Candidate thresholds are conventions, and across GWAS and F_{ST} thresholds, the same four genes were recovered, with L-homoserine biosynthesis ranked first or second among 468 pathways. It became significant after correction when stricter sets reduced the number of background genes (Supplemental Figure S14), so its consistent ranking is more informative than any single P value. Drift could still contribute to the F_{ST} candidate regions given $N_e \approx 25$–30 (Supplemental Figure S13), and soil N candidates may reflect local adaptation even though they passed tests for geography and climate. For the two soil-N candidates, the minor alleles came from many independent collection sites and were more frequent at N-rich sites, most clearly for ω-amidase within Mexico (Supplemental Figure S8), so neither signal rests on a single site, although shared ancestry may contribute.

Therefore, these genes are strong candidates for direct tests, such as comparing Asp-family amino acids among lines with alternative alleles at *Zm00001eb405410*, *Zm00001eb220040*, *Zm00001eb214360*, and *Zm00001eb094670*. Aspartate is a branch point in plant N metabolism. It is the precursor of Asn, a major transport and storage form of N (Lea et al., 2007). It also exchanges amino groups with Glu via transamination, and it is synthesized from the TCA cycle intermediate oxaloacetate, linking amino-acid N to carbon metabolism (Han et al., 2021). Variation at the entry to the Asp-family pathway could shift how Asp-derived N is allocated among Asn, Asp-family amino acids, and the Glu/Gln pool, consistent with the association of an aspartate kinase allele with soil N in landraces. Links from Asp to Glu/Gln and the ω-amidase pathway are discussed next.

Candidate genes at the aspartate–glutamate junction link amino acid N with C–N recycling

Glutamate exchanges amino groups with the TCA-cycle intermediate 2-oxoglutarate, which provides the carbon skeleton for ammonium assimilation through the GS/GOGAT cycle and glutamate dehydrogenase, and whose level signals C and N status in many organisms (Huergo and Dixon, 2015). Together, Asp, Asn, Glu, and Gln make up most of the free amino acid pool and are major N transport forms in plants (McAllister et al., 2016). Because no Glu/Gln pathway recurred across analyses, we discuss these junction candidates individually.

Each analysis implicated different genes at the junction. On the Asp side, Jarvis regions contained the aspartate aminotransferase *Zm00001eb152450* (GOT1), which interconverts Asp and 2-oxoglutarate with oxaloacetate and Glu (Han et al., 2021; Zhou et al., 2009). This gene was recently identified as *THP3*, whose teosinte alleles improve N assimilation and seed protein in maize (Huang et al., 2026). In contrast, overexpressing a cytosolic aspartate

aminotransferase from alfalfa in *Brassica napus* did not increase N use efficiency (McAllister et al., 2016), suggesting that the effects of this enzyme depend on the isoform and the tissue in which it is expressed. The soil-N GWAS identified an ω-amidase (*Zm00001eb005710*) that completes Asn/Gln transaminase routes by returning carbon to oxaloacetate and 2-oxoglutarate while releasing ammonia (Zhang and Marsolais, 2014).

On the Glu side, the clearest case is Indian Chief, the only population whose free amino acid profile changed (Figure 3). Most of the Cycle 14 increase in total free amino acids was Pro, so the lower Asp and Glu fractions largely reflect Pro accumulation. Indian Chief regions included two glutamate dehydrogenases (*Zm00001eb060520*, *Zm00001eb162300*), a glutamine synthetase (*Zm00001eb162900*), and a class I glutamine amidotransferase (*Zm00001eb217220*) that was also a candidate for free Pro in the amino acid GWAS, but no genes for Pro synthesis. GDH mutants show altered amino acid profiles under carbon limitation (Miyashita and Good, 2008; Dellero, 2020), and cereal GS supplies Gln to developing grain (Wei et al., 2021; Fujita et al., 2022). Because Pro is synthesized from Glu (**??**), increased flux from Glu to Pro, rather than increased primary N assimilation, is a plausible explanation, although we did not measure flux or enzyme activity. Glutamine amidotransferases function broadly, so *Zm00001eb217220* cannot be assigned to the Glu/Gln cycle without functional evidence, although rice *FLOURY ENDOSPERM19* affects grain filling and C–N metabolism (Lou et al., 2021). Because most grain N is protein-bound, these shifts changed the soluble pool without affecting total grain amino acid N. Jarvis showed no corrected change, consistent with Moll et al. (1994), in which its grain N gain came mainly from higher grain weight and post-anthesis N uptake. In the amino acid GWAS, GS/GOGAT genes (*Zm00001eb329710*, *Zm00001eb360480*, *Zm00001eb059870*) were associated with ratio traits, including Asn/Glu. Thus, all three populations implicated genes at the junction where Asp and Glu exchange N with TCA-cycle carbon. These genes differ among analyses and lack pathway-level support, but finding the same junction in three independent designs makes it a priority for testing how maize balances amino acid N against carbon supply.

Amino acid transport and protein turnover candidates

Beyond Asp and Glu metabolism, each analysis contained candidates for amino acid transport or protein turnover, and these differed among populations in ways that matched their phenotypes. They are individual candidates rather than recurring pathways, and because F_{ST} windows in small populations can reflect drift (Gerke et al., 2015), we considered Indian Chief and Jarvis candidates only when they matched the amino acid phenotypes.

Protein-turnover candidates were most relevant in Indian Chief, where Cycle 14 grain had higher free amino acids but unchanged protein-bound amino acids. Candidate regions included *Zm00001eb352300*, homologous to Arabidopsis *ATG18a* required for autophagosome formation during nutrient stress and senescence (Xiong et al., 2005), and the cysteine protease *SAG39* (*Zm00001eb214780*). Its Arabidopsis homolog *SAG12* localizes to senescence-associated vacuoles (Otegui et al., 2005) and supports N allocation to seeds (James et al., 2018). The same window also contained *Zm00001eb214890*, homologous to the ammonium transporter *AtAMT2* (Neuhäuser et al., 2009). Increased protease or autophagy activity in Cycle 14 could expand the free amino acid pool, but these activities were not measured.

Jarvis candidate regions instead contained genes involved in amino acid import and N signaling. An LHT-like gene (*Zm00001eb132750*) encodes a plasma membrane amino acid importer. In Arabidopsis, LHT1 imports amino acids into root and leaf cells, reducing losses (Fan et al., 2023). A ProT-like gene (*Zm00001eb159580*) transports Pro, glycine betaine, and GABA between cells (Lehmann et al., 2011). These regions also included a tandem pair of CC-type glutaredoxin genes (*Zm00001eb131830*, *Zm00001eb131840*) in two highly differentiated windows (ranked 2nd and 7th out of 2,151). In a reanalysis of public maize leaf RNA-seq data (Mu et al., 2017; Guo et al., 2022; Ma et al., 2021; Torres-Rodríguez et al., 2021; Alonso-Nieves et al., 2022; Wang et al., 2025), both genes were strongly downregulated under low nitrate (*Zm00001eb131840*; about 80-fold lower), although their expression in developing kernels is unknown (Supplemental Table S22). In Arabidopsis, CC-type glutaredoxins function in N signaling, including nitrate-induced regulators of root growth and the mobile peptides CEPD1/CEPD2 that increase nitrate uptake under N demand (Patterson et al., 2016; Ohkubo et al., 2017). Both glutaredoxin genes and the LHT-like gene also lie within the chromosome 3 Asp and Pro QTL interval of the B64 × CML14 population (Supplemental Figure S5). Because Jarvis increased grain N mainly via post-anthesis N uptake (Moll et al., 1994), amino acid import and N-signaling genes may be more relevant than changes in grain amino acid composition. Jarvis regions also contained NPL4-like protein 1 (*Zm00001eb146560*). Its Arabidopsis homolog partners with UFD1 and CDC48 to target ubiquitinated proteins (including Rubisco large subunit) for proteasomal degradation (**?**).

In the amino acid GWAS, several transporter candidates mapped to specific free amino acids. Three UmamiT genes (WAT1-related exporters) were associated with Gln (*Zm00001eb379220*), Asp (*Zm00001eb207630*), and Lys (*Zm00001eb352400*). Because seed-coat UmamiTs export maternal amino acids to the embryo and mutants reduce supply (Müller et al., 2015), these associations suggest that variation in export to the grain can alter individual free amino acids without changing synthesis. The NRT1/PTR family gene *NPF4.6* (*Zm00001eb277160*), whose family transports nitrate, peptides, amino acids, and hormones (Léran et al., 2014), was linked to free Lys. The Gln/Glu ratio mapped to the amino acid permease *aaap40* (*Zm00001eb261470*) in a region also containing *LHT1* and *LHT2*, while a cationic amino acid transporter (*Zm00001eb050630*) was associated with branched-chain amino acids. Two loci pointed to vacuolar storage, including an *AVT3A*-like gene (*Zm00001eb429100*) associated with Ile, consistent with Arabidopsis homologs that export vacuolar amino acids (Fujiki et al., 2017), and *AVT6C* (*Zm00001eb368680*) associated with free Asn. Thus, transport and turnover candidates differed across populations in phenotype-consistent ways: protein turnover in Indian Chief; amino acid import and N signaling in Jarvis; and amino acid export and vacuolar storage in the amino acid GWAS. Collectively, these candidates trace N flow to the grain—from uptake and recycling to delivery and storage—suggesting each population varies at a different step.

Conclusion

Across three maize populations, candidate genes for grain amino acid variation fell into three groups. The first is the aspartate-to-homoserine pathway, the only pathway significant after correction, in which each analysis implicated a different entry step, aspartate kinase in Jarvis (*Zm00001eb094670*) and the landrace soil-N GWAS (*Zm00001eb220040*), aspartate-semialdehyde dehydrogenase in Indian Chief (*Zm00001eb214360*), and homoserine dehydrogenase in the Goodman–Buckler panel (*Zm00001eb405410*). These steps may control how Asp-derived N is divided between Asn and the Asp-family amino acids. The second is the junction where Asp and Glu exchange N with TCA-cycle carbon, with aspartate aminotransferase in Jarvis, ω-amidase in the landraces, and glutamate dehydrogenase and glutamine synthetase in Indian Chief. The third is amino acid transport and protein turnover, with a peptidase A1 gene (*Zm00001eb190800*) at a Goodman–Buckler locus affecting amino acids from four biosynthetic families, senescence protease and autophagy candidates in Indian Chief, amino acid importers in Jarvis, and UmamiT exporters and vacuolar AVT transporters linked to specific free amino acids in the Goodman–Buckler panel.

These genes are candidates, not confirmed mechanisms. Pathway recurrence was significant after correction only with stricter candidate sets, drift may underlie the differentiation between cycles, and we did not measure N uptake or remobilization. Because the three groups act sequentially, partitioning N among amino acids, linking amino acid N to carbon supply, and reallocating N between tissues and storage, they can be tested together. For example, by combining alleles of an Asp-pathway gene and an amino acid transporter in the same background. Testing alleles and mutants, along with kernel expression under varying N and ^{15}N labeling to distinguish uptake from remobilization, will identify which steps, alone or combined, are best targets to improve maize N use.

Materials and methods

Plant material

Goodman–Buckler maize association panel

The Goodman–Buckler maize association panel (Flint-Garcia et al., 2005) was grown at the Genetics Farm near Columbia, Missouri, in 2017 and 2018, with two replicates per year in a randomized complete block design. Of the 282 lines in the panel, 280 were grown successfully. Thirteen kernels of each line were planted in a 10-foot row and self-pollinated. At physiological maturity, ears were harvested and husked, and the ears were dried at 40 °C and <20% relative humidity for 5 days. Ears from each line were shelled and bulked within each replicate and year, giving four composite grain samples per line. For amino acid analysis, 25 random kernels from each bulked sample were ground and prepared for extraction and quantification.

B64 × CML14 mapping population

The B64 × CML14 mapping population was developed by crossing B64 and CML14, the Goodman–Buckler lines with the lowest and highest grain free Asn, respectively. Field design, harvest, and sample preparation for the $F_{2:3}$ families are described in Fitzsimmons et al. (2026).

Maize landraces

We used 3,107 georeferenced maize landrace accessions from the CIMMYT International Germplasm Bank (Romero Navarro et al., 2017), collected in 28 countries and territories across Mesoamerica (2,296), South America (667), and the Caribbean (144) for which environmental variable such as soil N values were available at the collection sites.

Indian Chief and Jarvis recurrent-selection populations

We used two open-pollinated populations, Indian Chief and Jarvis Golden Prolific, from a program of reciprocal recurrent selection for grain yield (Moll and Stuber, 1971; Moll et al., 1994), sampling the base populations (Cycle 0) and the populations after 14 cycles of selection (Cycle 14). Because parents in each population were selected on the yield of their testcrosses with the other population, allele-frequency changes in the two populations are not fully independent. However, the testcrosses were used only for evaluation, and each new cycle was formed by intermating selected plants within the same population, so no alleles were exchanged between the populations.

For amino acid profiling, Indian Chief and Jarvis Cycle 0 and Cycle 14 plants were grown in 2025 at the Central Crops Research Station (Clayton, NC) under standard field management in a randomized complete block design. Whole cobs with fully developed kernels were harvested and stored until analysis.

Phenotyping

Amino acid profiling

PBAAs were extracted and quantified as described in Yobi and Angelovici (2018). This protocol utilizes microscale protein hydrolysis and absolute quantification using multiple reaction monitoring-based liquid chromatography–tandem mass spectrometry detection. During acid hydrolysis, Gln and Asn are converted into Glu and Asp, respectively. Therefore, we denote both Gln and Glu as Glx and Asn and Asp as Asx. During the extraction, Trp and Cys were fully destroyed, and Gly was poorly detected, so these three amino acids were excluded from the dataset. In total, 15 PBAAs were analyzed in the Goodman–Buckler panel, with Asx and Glx each representing two amino acids. In Indian Chief and Jarvis, Gly was also quantified, giving 16 PBAAs. FAAs were extracted and quantified as described in Gu et al. (2007). Unlike the protocol above, all 20 proteinogenic amino acids were independently and successfully detected using UPLC-MS/MS.

Amino acid-derived N was estimated by weighting each amino acid by its number of N atoms (4 for Arg, 3 for His, 2 for Asn, Gln, Lys, and Trp, and 1 for the others). Protein-bound Asx and Glx, which combine Asp with Asn and Glu with Gln after hydrolysis, were assigned 1.5 N. For the Goodman–Buckler GWAS trait *Total_N*, amino acid measurements were converted to amino acid-derived nitrogen values by weighting each amino acid by its nitrogen content. For an amino acid X, the nitrogen mass fraction was calculated as

$$\%N_X = \left(\frac{n_{N,X} \times M_N}{M_X}\right) \times 100$$

where M_X is the molar mass of amino acid X, M_N is the molar mass of nitrogen, and $n_{N,X}$ is the number of nitrogen atoms in amino acid X. Amino acid-derived nitrogen for each amino acid was then calculated as

$$N_X\ (\mu g/mg) = \left(\frac{n_{N,X} \times M_N}{M_X}\right) \times AA_X\ (\mu g/mg)$$

and summed across amino acids before BLUP estimation and GWAS.

Free and protein-bound amino acid pools

Free and protein-bound pools were compared in the 279 Goodman–Buckler lines with both measurements (Shrestha et al., 2022) using paired Wilcoxon signed-rank tests. Amino acid-derived N was calculated as described above. To compare composition on the same basis, free Asp and Asn and free Glu and Gln were combined to match Asx and Glx, and Gly, Trp, and Cys were excluded from the free-pool total.

Indian Chief and Jarvis amino acid comparisons

Technical replicates were averaged within each biological sample (Indian Chief Cycle 0 $n = 5$, Cycle 14 $n = 8$, Jarvis Cycle 0 $n = 3$, Cycle 14 $n = 8$). For each sample we calculated total FAA and PBAA per mg and each amino acid's fraction within its pool. Amino acid-derived N per mg was calculated as described above, and total amino acid-derived N was the sum of FAA- and PBAA-derived N, estimating N in amino acids (not total grain N). Cycle 0 and Cycle 14 were compared within each population with two-sided Wilcoxon rank-sum tests (normal approximation), with Cliff's δ as the effect size. P values were Benjamini–Hochberg adjusted within each of four test families, the eight FAA metrics (Supplemental Table S14), the amino acids within each pool (Supplemental Table S15), the PBAA metrics (Supplemental Table S16), and the three amino acid-derived N measures (Supplemental Table S17). With three Cycle 0 Jarvis samples, the minimum possible P was 0.019.

Soil nitrogen

The environmental phenotype, soil total N, was taken from SoilGrids 2.0 (Poggio et al., 2021) at 250 m resolution. Values for the 0–5 cm and 5–15 cm layers were assigned to the collection coordinates of each landrace accession, and 0–5 cm was used as the primary GWAS phenotype and 5–15 cm as a check.

Genotyping

Goodman–Buckler panel

The Goodman–Buckler panel was genotyped with the maize HapMap 3.2.1 SNPs for the 282-line panel (Bukowski et al., 2018), 4,177,796 SNPs positioned on the B73 AGPv4 assembly. Positions were lifted to B73 RefGen_v5 (Zm-B73-REFERENCE-NAM-5.0), referred to below as reference genome version 5, as described under Amino acid GWAS.

B64 × CML14 mapping population

F_2 plants were genotyped with the mid-density temperate corn panel of Agriplex Genomics (Cleveland, OH) as described in Fitzsimmons et al. (2026).

Maize landraces

The landraces were genotyped by genotyping-by-sequencing at about 2× coverage, with SNP calling and imputation by Romero Navarro et al. (2017). We used 495,755 SNPs positioned on reference genome version 5 for the 3,107 accessions with soil N values.

Indian Chief and Jarvis

Each population–cycle group included 96 individuals (384 in total). Paired-end Illumina libraries were prepared with Twist Bioscience kits and sequenced as 2 × 150 bp reads on a NovaSeq 6000, and reads were demultiplexed with `fqtk` v0.3.1.

Reads were trimmed/filtered with `fastp` v0.23.4, QC-checked with `FastQC` v0.12.1, aligned to RefGen_v5 (GCF_902167145.1) using `BWA-MEM` v0.7.18, and deduplicated with `Picard MarkDuplicates` v3.1.1. Variants were jointly called with GATK v4.5.0.0 Best Practices (`CombineGVCFs`, `GenotypeGVCFs`) and hard-filtered, yielding 52,543,238 SNPs. After removing five low-quality samples, 379 individuals remained. For population structure, we kept biallelic SNPs on the 10 chromosomes with QUAL $\geq$ 30, MAF $\geq$ 0.01, site depth 221–413, and data in $\geq$50% of individuals (`bcftools` v1.18), leaving 512,667 SNPs. Mean depth at these SNPs was 0.73× per individual (0.32–1.91×); genome-wide coverage including zero-read sites was 0.28–0.38× per group (`Picard CollectWgsMetrics`), consistent with low-pass skim sequencing.

Heritability and consistency across years

Broad-sense heritability (H^2) on a line mean basis was estimated on the log scale from the plot records (two years × two blocks) with the linear mixed model $y \sim 1 + (1|\text{taxa}) + (1|\text{year}) + (1|\text{taxa:year}) + (1|\text{year:block})$ for 40 free and 76 protein-bound amino acid traits, and the Cullis generalized H^2 was computed from the same model. Genomic heritability (h^2) was estimated with a VanRaden genomic relationship matrix for the 272 genotyped lines, with 95% profile-likelihood intervals. Each trait was also fitted with a permuted phenotype, and traits for which the permuted phenotype returned a non-zero h^2 were excluded from comparisons of h^2 with H^2. Consistency across years was measured as the Spearman correlation between 2017 and 2018 line means for the 251 lines grown in both years. For soil N in the landrace panel, genomic heritability was estimated for the 3,107 accessions before and after removing 10 and 50 genotype principal components from the phenotype, and compared with 1,000 permuted phenotypes.

Genome-wide association studies

Amino acid GWAS

We performed GWAS for 78 amino acid phenotypes (Supplemental Table S2) using phenotype BLUPs and 4,177,796 HapMap 3.2.1 SNPs in the HAPPI pipeline (Slaten et al., 2020), applying GAPIT MLM, MLMM, BLINK, and FarmCPU. Significance was set at 5% Benjamini–Hochberg FDR, calculated separately for each phenotype and model. Because MLMM yielded no significant SNPs, we report results from MLM, BLINK, and FarmCPU. SNP coordinates were lifted from RefGen_v4 to RefGen_v5 using the B73 chain file, Twenty one of 2,461 significant SNPs in alignment gaps were moved to the nearest aligned base. Candidate genes were RefGen_v5 genes within 25 kb of significant SNPs.

Soil N GWAS and environmental specificity tests

GWAS of soil N used the 3,107 landrace accessions and 495,755 SNPs, with MLM, BLINK, and FarmCPU (Slaten et al., 2020) and three genotype principal components as covariates. Significance was declared at a 5% Benjamini–Hochberg FDR computed over the full scan for each model, and candidate genes were RefGen_v5 genes within 25 kb of a significant SNP.

To assess whether soil-N associations reflected geography or climate, we repeated the GWAS for latitude, longitude, elevation (Romero Navarro et al., 2017), soil pH, and CEC (0–5 cm) (Poggio et al., 2021), and aridity (Zomer et al., 2022). For each variable pair, we tested overlap of significant SNPs against chance ($n_1 n_2 / M$, M = 495,755) with one-sided hypergeometric tests. Independently, we ranked all SNPs by $|r|$ between allele dosage and each site variable across 3,107 accessions. A soil N SNP was classified as environment linked if it was in the top 1% for any non-N variable, not linked if it was in the top 5% for soil N but not in the top 1% for any other variable, and inconclusive otherwise. We compared the fraction of soil N SNPs in the top 1% for any geographic/environmental variable by binomial tests against the whole genome and against SNPs in the top 1% for soil N.

For the two focal soil-N SNPs (S1_16128632, S5_22338979), we compared soil N among genotypes with Kruskal–Wallis tests (ε^2), followed by Dunn tests with Holm correction, and compared minor-allele carriers vs non-carriers within Mexico and within Central America using Wilcoxon rank-sum tests. These tests do not account for population structure.

QTL mapping

The genetic map of 610 markers was constructed with R/qtl (Broman et al., 2003) as described in Fitzsimmons et al. (2026). Best linear unbiased predictions (BLUPs) of free Asp and free Pro were calculated for the 250 $F_{2:3}$ families. QTL scans used Haley–Knott regression, significance thresholds were set at $\alpha = 0.05$ from 1,000 permutations, and support intervals were defined as 1.5-LOD drops from the QTL peak.

Genes within the shared 1.5-LOD support interval (chr3:14.37–143.31 Mb) were taken from the reference genome version 5 gene annotation in MaizeGDB (Woodhouse et al., 2021a), with 192 non-coding genes absent from that annotation added from Ensembl Plants release 62. Genes were annotated with gene name, PANTHER family, protein class, and Gene Ontology terms, and with kernel expression from qTeller (Woodhouse et al., 2021b), which provides RNA-seq transcript abundance across maize kernel tissues and developmental stages (Supplemental Table S8).

Population-genomic analyses of Indian Chief and Jarvis

Genotype likelihoods and differentiation between cycles

Population-genomic statistics were estimated from genotype likelihoods in ANGSD v0.940 (Korneliussen et al., 2014) for each population–cycle group using identical filters. Per-chromosome site allele-frequency likelihoods (SAF) were computed with the SAMtools genotype-likelihood model from properly paired, unflagged reads (mapping quality ≥ 30), bases (quality ≥ 20), and sites with total depth ≥ 2. The reference genome version 5 allele was set as the major allele, and analyses were restricted to polymorphic sites within each group (SNP likelihood-ratio $P < 10^{-6}$, minor allele frequency ≥ 0.0005, triallelic sites removed).

Allele-frequency change from Cycle 0 to Cycle 14 was quantified separately in Indian Chief and Jarvis using F_{ST}. For each contrast, a joint 2D site-frequency spectrum was estimated from two SAF files with `winsfs` (Rasmussen et al., 2022) and used as the prior for `realSFS fst`. Per-site F_{ST} was smoothed along chromosomes with GenWin (spline smoothness 500 kb), which sets window boundaries at spline inflection points (Beissinger et al., 2015). Each window was summarized by mean per-site F_{ST}. Windows at or above the 95th percentile of each population's window-F_{ST} distribution were retained as candidate regions. These are empirical outliers, not significance thresholds, because drift at these population sizes shifts linked regions together.

To compare candidate regions with different window boundaries, we split the genome at any boundary from either population and assigned each segment the F_{ST} of the Indian Chief and Jarvis windows containing it. Overlap was the number of base pairs in the top 5% in both populations, assessed against 10,000 circular shifts of the Jarvis track along concatenated chromosomes to preserve window size and clustering. Concordance in window F_{ST} was summarized by Spearman's correlation across segments.

Population structure and differentiation between Indian Chief and Jarvis

Because sequencing depth varied among cycles, we inferred population structure from genotype likelihoods rather than called genotypes. From the filtered VCF, we retained sites with reads in at least half of the 379 individuals and minor allele frequency ≥ 0.05, then thinned to one site per 10 kb to reduce linkage (82,061 sites). We estimated admixture with NGSadmix (Skotte et al., 2013) for K = 2–5 (three runs per K) using the 12,031 sites passing its MAF filter, and chose the largest K for which runs converged. We computed principal components from the genotype-likelihood covariance matrix with PCAngsd (Meisner and Albrechtsen, 2018). We measured differentiation between Indian Chief and Jarvis at Cycle 0 and Cycle 14 using Hudson's F_{ST} (Bhatia et al., 2013), calculated genome-wide and per chromosome from joint site-frequency spectra estimated with `winsfs` (Rasmussen et al., 2022) from ANGSD SAF files.

Effective population size

Because each cycle was formed by intermating the selfed progenies of 20 selected plants within each population (Moll and Stuber, 1971), neutral allele-frequency change is expected to follow a Wright–Fisher process with one generation per cycle and an effective population size (N_e) of at most 20, the value reached when all selected parents contribute equally.

To compare the observed differentiation with this expectation, we estimated N_e for each population from the allele-frequency change between Cycle 0 and Cycle 14. Per-site allele frequencies were taken from ANGSD (Korneliussen et al., 2014), retaining sites with data for at least 30 individuals and the same major and minor alleles in both cycles. Sites were grouped into non-overlapping 250-kb windows (at most 400 randomly chosen sites per window, 8,524 windows in Indian Chief and 8,527 in Jarvis), Hudson's F_{ST} (Bhatia et al., 2013) between cycles was computed per window as a ratio of sums, and the genome-wide mean was taken across windows. We then simulated neutral Wright–Fisher drift for 14 generations from the observed Cycle 0 frequencies for N_e = 20, 30, 50, 75, 100, 150, 250, and 500, resampled the simulated frequencies at both time points using the number of individuals with data at each site, and computed F_{ST} with the same estimator. N_e was taken as the value at which the simulated genome-wide mean F_{ST} matched the observed value, interpolating linearly between grid points.

Candidate window annotation

Candidate genes were annotated with PANTHER family and protein class (Thomas et al., 2022) and Gene Ontology terms (Gene Ontology Consortium et al., 2026) (Supplemental Tables S20–S21).

Pathway recurrence across analyses

Pathway membership was obtained from CornCyc (Plant Metabolic Network release 2023, Hawkins et al., 2021). For the main test, we used Bonferroni-significant GWAS SNPs ±25 kb ($P < 0.05/4{,}177{,}796$ for amino acids and $P < 0.05/495{,}755$ for soil N) and the top 3% of windows from each F_{ST} scan. A pathway was considered a hit if any pathway gene overlapped a set interval, and we recorded how many sets hit each pathway. The null distribution was generated from 200,000 permutations by circularly shifting each set's intervals along each chromosome by an independent random offset, preserving interval number, size, spacing, and clustering. For each pathway, P was the fraction of permutations with at least as many set hits as observed (permutation-based empirical P). We corrected for testing all 468 pathways with a minimum-P procedure, using one half of the permutations to estimate each pathway's tail probabilities and the other half to obtain the distribution of the smallest P per permutation. To assess sensitivity to thresholds, we repeated the test for 12 combinations of GWAS thresholds (Benjamini–Hochberg FDR 5%, 3%, and 1%, computed per model over the full scan, and Bonferroni) and F_{ST} window thresholds (top 5%, 3%, and 1%), with 20,000 permutations per combination (200,000 for the Bonferroni sets with the top 5% and 3% of windows).

Expression of candidate genes under contrasting N supply

Expression responses of candidate genes to N supply were taken from a reanalysis of published maize RNA-seq studies of root, leaf, shoot, and whole-seedling tissue, listed with their DOIs in Supplemental Table S22, including six leaf N-deficiency studies (Mu et al., 2017; Guo et al., 2022; Ma et al., 2021; Torres-Rodríguez et al., 2021; Alonso-Nieves et al., 2022; Wang et al., 2025). Reads from each study were quantified against reference genome version 5 with salmon in the nf-core/rnaseq pipeline, and each study was analyzed with DESeq2, testing genes with at least 10 reads in at least two samples. Within each study, contrasts of the same tissue and contrast type were combined by inverse variance, and studies were then pooled with DerSimonian–Laird random-effects meta-analysis. Only replicated contrasts were pooled, and P values were adjusted by the Benjamini–Hochberg method within each tissue and contrast type.

Accession numbers

Sequence data from this article can be found in the NCBI Sequence Read Archive under BioProject PRJNAXXXXXX. Gene identifiers refer to the B73 RefGen_v5 annotation (Zm00001eb).

Supplemental data

The following materials are available in the online version of this article.

Supplemental Figure S1. Free amino acid composition of the Goodman–Buckler panel.

Supplemental Figure S2. Heritability of grain amino acid phenotypes in the Goodman–Buckler panel.

Supplemental Figure S3. Free (FAA) and protein-bound (PBAA) amino acid pools across matched Goodman–Buckler lines ($n = 279$).

Supplemental Figure S4. Manhattan plots for a locus on chromosome 9 associated with nine amino acid phenotypes.

Supplemental Figure S5. Free Asp and Pro map to the same QTL on chromosome 3 in the B64 × CML14 population.

Supplemental Figure S6. Overlap of soil-N associations with other environmental GWAS.

Supplemental Figure S7. Genotype–environment correlation test for the soil-N SNPs.

Supplemental Figure S8. Geography and soil N of the two focal soil-N SNPs.

Supplemental Figure S9. Protein-bound amino acid (PBAA) composition in Indian Chief and Jarvis at Cycle 0 (C0) and Cycle 14 (C14).

Supplemental Figure S10. Amino acid-derived nitrogen per mg sample in Indian Chief and Jarvis at Cycle 0 (C0) and Cycle 14 (C14).

Supplemental Figure S11. Per-sample sequencing depth in the Indian Chief and Jarvis skim-sequencing data.

Supplemental Figure S12. Quality summaries for the 512,667 filtered SNPs.

Supplemental Figure S13. Effective population size of the Indian Chief and Jarvis recurrent-selection populations.

Supplemental Figure S14. Sensitivity of the pathway recurrence test to candidate thresholds.

Supplemental Table S1. Free amino acid phenotypes across the Goodman–Buckler Association Panel.

Supplemental Table S2. Amino acid trait definitions.

Supplemental Table S3. Abundance, variability, and heritability of the 20 free amino acids.

Supplemental Table S4. Heritability of free and protein-bound amino acid traits.

Supplemental Table S5. Free vs. protein-bound amino acid pools.

Supplemental Table S6. Amino acid GWAS candidate genes.

Supplemental Table S7. Amino acid GWAS loci.

Supplemental Table S8. Genes in the chromosome 3 Asp/Pro QTL interval of the B64 × CML14 population.

Supplemental Table S9. Soil N GWAS candidate genes.

Supplemental Table S10. Environmental GWAS overlap test.

Supplemental Table S11. Soil-N GWAS SNPs and their overlap with other environmental GWAS.

Supplemental Table S12. Genotype–environment correlation test for soil-N SNPs.

Supplemental Table S13. Expectation test for the genotype–environment correlation.

Supplemental Table S14. Free amino acid metrics in Indian Chief and Jarvis, Cycle 0 vs. Cycle 14.

Supplemental Table S15. Amino acid composition in Indian Chief and Jarvis, Cycle 0 vs. Cycle 14.

Supplemental Table S16. Protein-bound amino acid metrics in Indian Chief and Jarvis, Cycle 0 vs. Cycle 14.

Supplemental Table S17. Amino acid-derived N per mg sample in Indian Chief and Jarvis.

Supplemental Table S18. Indian Chief F_{ST} windows, Cycle 0 vs. Cycle 14.

Supplemental Table S19. Jarvis F_{ST} windows, Cycle 0 vs. Cycle 14.

Supplemental Table S20. Indian Chief candidate genes in top-5% F_{ST} windows.

Supplemental Table S21. Jarvis candidate genes in top-5% F_{ST} windows.

Supplemental Table S22. Expression responses of candidate genes to N supply in public maize RNA-seq.

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Author contributions

TODO.

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Conflict of interest statement

None declared.

Data availability

Raw sequencing reads for the Indian Chief and Jarvis populations are available from the NCBI Sequence Read Archive under BioProject PRJNAXXXXXX, and the filtered SNP set used for population structure analyses is available at Zenodo (doi:10.5281/zenodo.XXXXXXX). Amino acid phenotypes and all Supplemental Data are provided with this article. All code used in this study is available at `https://github.com/nirwan1265/Grain_N_remobilization` (archived at doi:10.5281/zenodo.XXXXXXX). The amino acid GWAS results for the Goodman–Buckler panel can be browsed through a Shiny web application at `https://j83gk2-nirwan-tandukar.shinyapps.io/amino_gwas_app/`

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